

Elecard StreamEye 2024

User Guide

User Guide Notices

Elecard StreamEye 2024 (4.10) User Guide

First edition: March 2013

Date modified: October 16, 2024.

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1. Introduction

1.1 Preface

Elecard StreamEye is a powerful software tool designed for professionals and prosumers in the video compression field. Elecard StreamEye enables the user to perform an effective in-depth analysis of video sequences.

Elecard StreamEye supports elementary video streams such as MPEG-1/2, AVC, HEVC, VVC, VP9, AV1, AVS3 which can be opened directly from MPEG-1 System Streams (SS), MPEG-2 Program Streams (PS), and MPEG-2 Transport Streams (TS), as well as from such popular container formats as AVI, MP4, MKV/WebM, FLV, MXF, IVF, HEIC, and MPEG-DASH MPD files.

There are two versions of Elecard StreamEye:

- StreamEye Basic which has limited functionalities and can be purchased as a standalone application.
- Full-featured StreamEye (or full version) which can be purchased as a standalone application or as a part of StreamEye Studio.

Automated analysis and standard compliance check are available with the command-line tool of the StreamEye full-featured version. Configuration via an XML file provides an easy and flexible way to pick specific information from the processed stream and save it into CSV files or into YUV RAW data in case of video data.

See a [comparison table](#) of basic and full-featured versions at the end of the document.

1.2 Describing Elecard StreamEye

1.2.1 Overview

Elecard StreamEye products enable the user to perform an effective in-depth analysis of video sequences. Elecard StreamEye provides a visual representation of the encoded video features and a stream structure analysis of MPEG-1 (ISO/IEC 11172-2), MPEG-2 (ISO/IEC 13818-2), AVC/H.264 (see ISO/IEC 14496-10), HEVC/H.265 (ISO/IEC 23008-2 MPEG-H Part 2), VVC/H.266 (ISO/IEC 23090-3), Google VP9 (see A VP9 Bitstream Overview), AVS3 (Information Technology Intelligent Media Coding (T/AI 109.2 — 2021) by Industry Alliance Standard), and AV1 (AV1 Bitstream & Decoding Process Specification) video.

With navigation down to the deepest levels of an encoded stream, StreamEye products enable powerful and efficient debugging for developing MPEG-1/2, AVC, HEVC, VVC, VP9, AV1, AVS3 video codecs. A detailed display of information includes frames' types and sizes, coded units' data, visualization of slice and tile boundaries, partition, motion vectors, types, bit sizes, quantizers, etc.

Note that the full-featured version of Elecard StreamEye contains all the above-mentioned features and provides more accurate configurations for the overlay and additional customizable options. Moreover, it allows viewing reference data and comparing the analysis results with it, calculating the selected metrics, using command line and other additional options and parameters that are not available in the Basic version.

1.2.2 Specifications

Elecard StreamEye supports the following formats (depends on the license purchased):

- MPEG-1 Video stream

- MPEG-2 Video stream
- AVC/H.264 Video stream
- HEVC/H.265 Video stream
- VVC/H.266 Video stream
- VP9 Video stream
- AV1 Video stream
- AVS3 Video stream
- MPEG-1 System stream (MPEG-1/2)
- MPEG-2 Transport Stream (MPEG-1/2, AVC, HEVC, VVC, AVS3)
- MPEG-2 Program Stream (MPEG-1/2, AVC, HEVC)
- MPEG-DASH MPD (AVC, HEVC)
- AVI file container (MPEG-1/2, AVC, HEVC)
- MP4 file container (MPEG-1/2, AVC, HEVC, VP9, AV1, VVC)
- MKV file container (MPEG-1/2, AVC, HEVC, VP9, AV1, AVS3)
- WebM file container based on MKV (VP9, AV1)
- IVF file container (VP9, AV1)
- FLV file container (AVC, HEVC)
- MXF file container (MPEG-2, AVC)
- HEIC file container (HEVC)

1.2.3 Features

Both versions of Elecard StreamEye provide the following features (depends on the license purchased):

- Displaying and saving stream summary and picture information.
- Displaying decoded, predicted, and unfiltered frame data (YUV or single component).
- Displaying residual, transform, and dequantized coefficients.
- Saving decoded, predicted, unfiltered, and residual information.
- Displaying Macroblock (MPEG-1/2, AVC) / Coding Tree Unit (HEVC, VVC) / Super Block (VP9, AV1) data: location, slice index, tile index, sizes, coded, prediction, transform unit info.
- Navigation via the BarChart.
- Stream navigation and display in I, P, B, IP, and Key Frames modes.
- Stream viewer to present file content (header level) in the text mode.
- Hex viewer.
- Possibility to view information about a part of a stream within the range specified by the user.

The full-featured version of Elecard StreamEye provides the following features (depends on the license purchased):

- Displaying bitstream headers with offsets and bit size indication (as in the Standard documentation) for all standards.
- Visualization of slice and tile boundaries, partition, motion vectors, types, bit sizes, quantizers.
- Navigation via the Thumbnails panel or the AreaChart.
- Displaying the bit distribution inside a video stream. Ability to choose bitstream elements to display.
- Displaying reference frames.
- Sharing comments between application instances and/or applications of the Elecard StreamEye Studio set.
- Synchronization between applications of the Elecard StreamEye Studio set (Binding mode).
- Displaying the Decoded Picture Buffer (DPB) information.

- Analysis and display of the Decoder Video Buffer parameters.
- Command-line tool.
- Viewing reference raw data.
- Metrics calculation.
- Visual comparison with reference raw data (Temperature, Subtraction, Compare, Block PSNR modes).
- EPSNR¹ calculation.
- Compliance verification of the stream against the compression standard (AVC and HEVC only).

Note 1: *EPSNR (Estimated Peak Signal-to-Noise Ratio) metric measures PSNR values from an encoded video sequence without a reference video sequence. Encoded stream data, including sequence and picture headers information, macroblock types, quantization parameters and other information from the stream, is used during the calculation. In almost all cases, this information is enough to retrieve PSNR values very precisely, though mismatches might happen in some rare situations.*

1.3 Using This Guide

1.3.1 Purpose

This guide is intended to help video encoder developers, SoC designers, and consumer electronics vendors to analyze video bitstreams (including MPEG-1/2, AVC/H.264, HEVC/H.265, VVC/H.266, AV1, AVS3, and VP9 streams).

1.3.2 Abbreviations and Terminology

The following section defines the terms used throughout this document.

Blocks:

- **CTU** (Coding Tree Unit) — a block used for HEVC and VVC formats.
- **Macroblock** — a block used for MPEG-1/2 and AVC.
- **Superblock** — a block used for VP9 and AV1.

Blocks are divided into subblocks:

- **CU** — coded unit
- **PU** — prediction unit
- **TU** — transform unit

Metric types:

- **PSNR** — Peak Signal-to-Noise Ratio.
- **APSNR** — average values of PSNR results calculated for a sequence and based on mean values of MSE results.
- **VQM** — Video Quality Measurement Techniques is a metric designed to calculate video quality and measure the perceptual effect of video distortions.
- **SSIM** — Structural SIMilarity. This metric is based on measuring three components (luminance similarity, contrast similarity and structural similarity) and combining them into result value.
- **DELTA** — The value of this metric is the mean difference of the color components in the correspondent points of image.
- **MSE** — Mean-squared-error based metric.
- **MSAD** — The value of this metric is the mean absolute difference of the color components in the correspondent points of the image.
- **VMAF** — Video Multimethod Assessment Fusion is an objective full-reference [video](#)

- [quality](#) metric developed by [Netflix](#) in cooperation with the [University of Southern California](#).
- **VMAF phone** — a metric designed to predict the quality of videos displayed on mobile phones.
 - **NQI** — New Quality Index is a metric designed to calculate video quality by combining three components: correlation loss, luminance and contrast distortion.
 - **VIF** — Visual Information Fidelity is a full reference image quality assessment index based on natural scene statistics and the notion of image information extracted by the human visual system.

Raw data:

- **I444, I422, IYUV, NV12, NV21, P420, P444, RGB24, RGB32, UYVY (Y422, UYNV, HDYC), V210, V400, V444, Y42B, YUY2 (YUNV, V422, YUYV), YV12, YV16, YVYU.**

For additional information about raw data types, follow the link: fourcc.org.

1.3.3 Topics Covered

The following lists the topics covered in this document:

- **Section 1: Introduction** provides a general overview of the Elecard StreamEye program and describes the purpose of the document and its contents.
- **Section 2: Getting Started** describes how to install, uninstall, and run the Elecard StreamEye program.
- **Section 3: Describing Elecard StreamEye** provides a detailed description of the Elecard StreamEye program, including features and supported stream formats.
- **Section 4: Using Elecard StreamEye** describes the Elecard StreamEye GUI.

1.3.4 Related Documentation

For additional information, review the following documents:

- ISO/IEC 11172-2 for MPEG-1 video
- ISO/IEC 13818-2 for MPEG-2 video
- ISO/IEC 14496-10 for AVC/H.264 video
- ISO/IEC 23008-2 MPEG-H Part 2
- ITU-T Recommendation H.265
- H.266 Versatile video coding
- A VP9 Bitstream Overview for Google VP9
- AV1 Bitstream & Decoding Process Specification
- AVS3 (Information Technology Intelligent Media Coding (T/AI 109.2 — 2021) by Industry Alliance Standard) System Requirements

1.4 System Requirements

1.4.1 Hardware Requirements

- 8 GB RAM (or higher).

1.4.2 Software Requirements

- macOS 10.13 High Sierra — Mac OS 14 Sonoma;

1.5 Licensing and Technical Support

By installing, copying, or otherwise using the software product or any updates, you agree to be bound by the terms of the "Elecard" End-User License Agreement ("EULA"). This EULA is a legal agreement between you (either an individual or a single entity) and Elecard for the "Elecard" software product(s) accompanying this EULA, which include(s) computer software and may include "online" or electronic documentation, associated media, and printed materials ("software product").

For technical support, please contact Elecard Technical Support Team: tsup@elecard.com

For sales and licensing information, contact Elecard Sales Department: sales@elecard.com

1.6 Activating Elecard StreamEye

1.6.1 Sentinel License Protection

Make sure that Sentinel License Manager software has been installed on your computer. License Manager is provided within the product installation pack for Windows OS and macOS and installed automatically.

License Manager is provided as a separate installation pack for Linux OS and should be installed manually.

Please note, macOS reinstallation will make the activated product license invalid.

To check that License Manager is properly installed, go to Sentinel Admin Control Center at <http://localhost:1947>. If you see your license information, the installation is successfully completed.

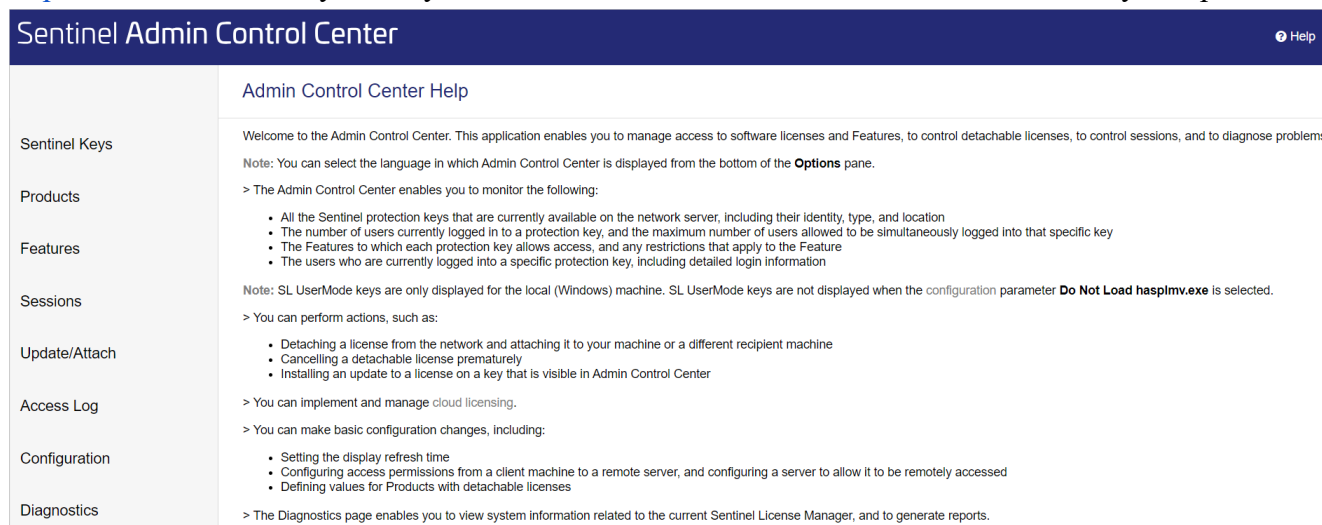


Figure 1. The License Manager with license information

License Manager allows configuring, controlling, monitoring and looking through a list of available licenses. For more details on License Manager operation click the *Help* tab.

Note, each page contains the Help tab related to this page only.

You can activate Elecard StreamEye offline via following steps:

For Elecard StreamEye activation, a C2V (client-to-vendor) file should be created. To create the C2V file follow the steps:

1. Open the *Application* tab (StreamEye) and select *Preferences* in the drop-down menu.
2. The *Options* dialog window should open. Open the *License* tab.

3. Select HASP-SL for PC-based or HASP-HL for dongle-based activation in the *Offline activation with license files* field. Note, if you select a PC-based (HASP-SL) activation, full-featured activated Elecard StreamEye will run only on this particular hardware. If you have Elecard dongle, plug it into a free USB port on your computer and select dongle-based (HASP-HL) activation. It will be possible to use Elecard StreamEye on any hardware with the dongle plugged in.

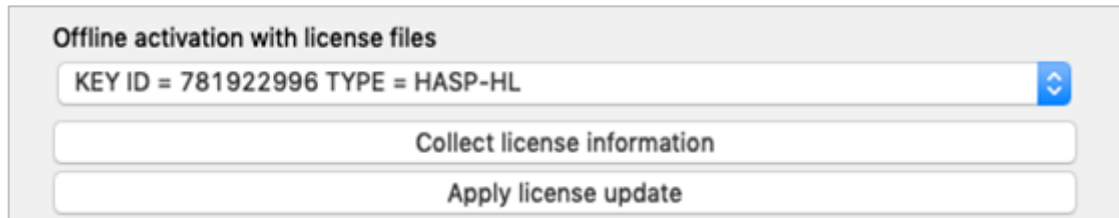


Figure 2. *Offline activation with license files*

4. Click the **Collect license information** button. The *Save File* dialog box should open. Click **Save** and save the C2V file to a required directory. Send the saved C2V file to Elecard Sales at sales@elecard.com.

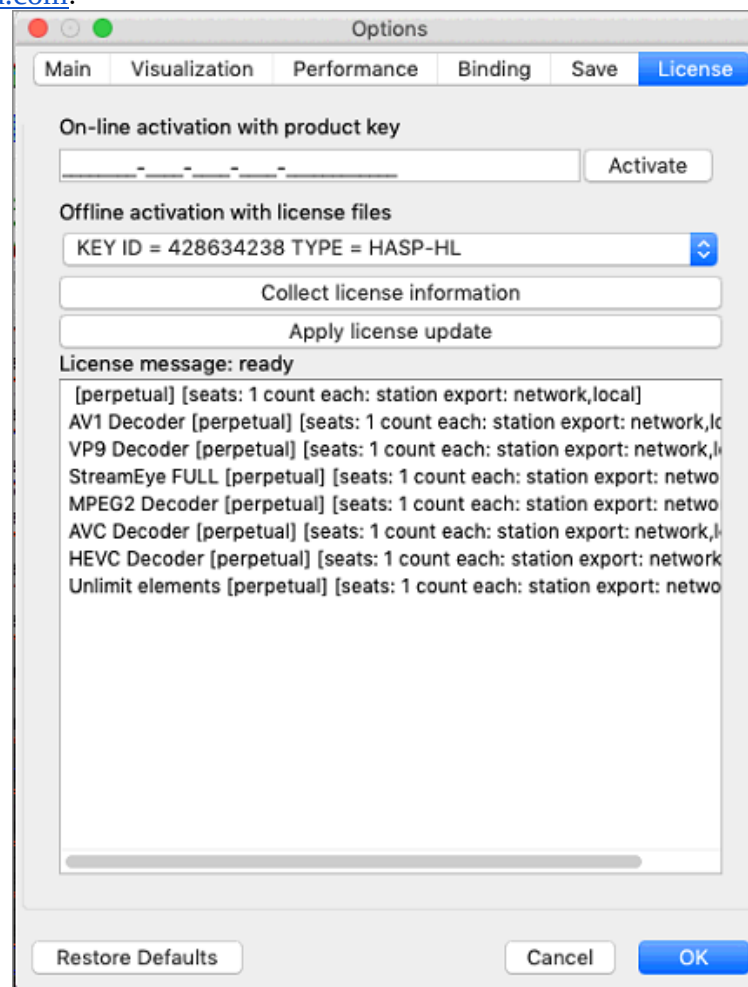


Figure 3. *Creating a C2V file: License tab*

5. After receiving the V2C file containing license information back from Elecard Sales, save the received file to a required directory. Open the *Application* tab and select *Preferences* in the drop-down menu again. Open the *License* tab in the *Options* dialog and click **Apply license update** (see Figure 4). The file selection dialog box should appear. Choose the received V2C file and click **OK**. Your application is successfully activated.

2. Getting Started

The following section details the procedures for installing and running Elecard StreamEye.

2.1 Installing Elecard StreamEye

Elecard StreamEye is supplied as part of Elecard StreamEye Studio or as a full-featured or basic standalone application. After installing Elecard StreamEye Studio or Elecard StreamEye on macOS, all applications are located in the Applications folder:

- *Applications* → *Elecard StreamEye Studio 2024*;
- *Applications* → *Elecard StreamEye 2024*.

The console is located in: *Users* → *bin* → *SEconsole* → *streameye-cli*.

The documents are located in: *Library* → *Application Support* → *Elecard* → *StreamEye 2024* → *Elecard StreamEye 2024.pdf*.

2.2 Uninstalling Elecard StreamEye

Elecard StreamEye can be uninstalled as part of Elecard StreamEye Studio or as a standalone application. To uninstall the application in macOS, move the required application to the trash icon.

Elecard StreamEye Studio or Elecard StreamEye applications are located in the Applications folder:

- *Applications* → *Elecard StreamEye Studio 2024*;
- *Applications* → *Elecard StreamEye 2024*.

The console is located in: *Users* → *bin* → *SEconsole*.

All documents are located in: *Library* → *Application Support* → *Elecard*.

2.3 Running Elecard StreamEye

The installed Elecard StreamEye Studio or Elecard StreamEye including all files packaged together are located in *Applications* → *Elecard StreamEye Studio 2024* or *Applications* → *Elecard StreamEye 2024*, and all consoles are located in *Users* → *bin* → *Elecard StreamEye Studio 2024*

To run Elecard StreamEye double click on the application icon, and the application will automatically run.

3. Using Elecard StreamEye

3.1 Introduction

The following section describes the Elecard StreamEye program GUI.

The Elecard StreamEye program has a simple user-friendly interface that allows the user to perform a number of operations including: opening media stream for analysis, visualization and saving various types of information, stream navigation, etc.

3.2 Describing Elecard StreamEye GUI

The following section describes the Elecard StreamEye GUI.

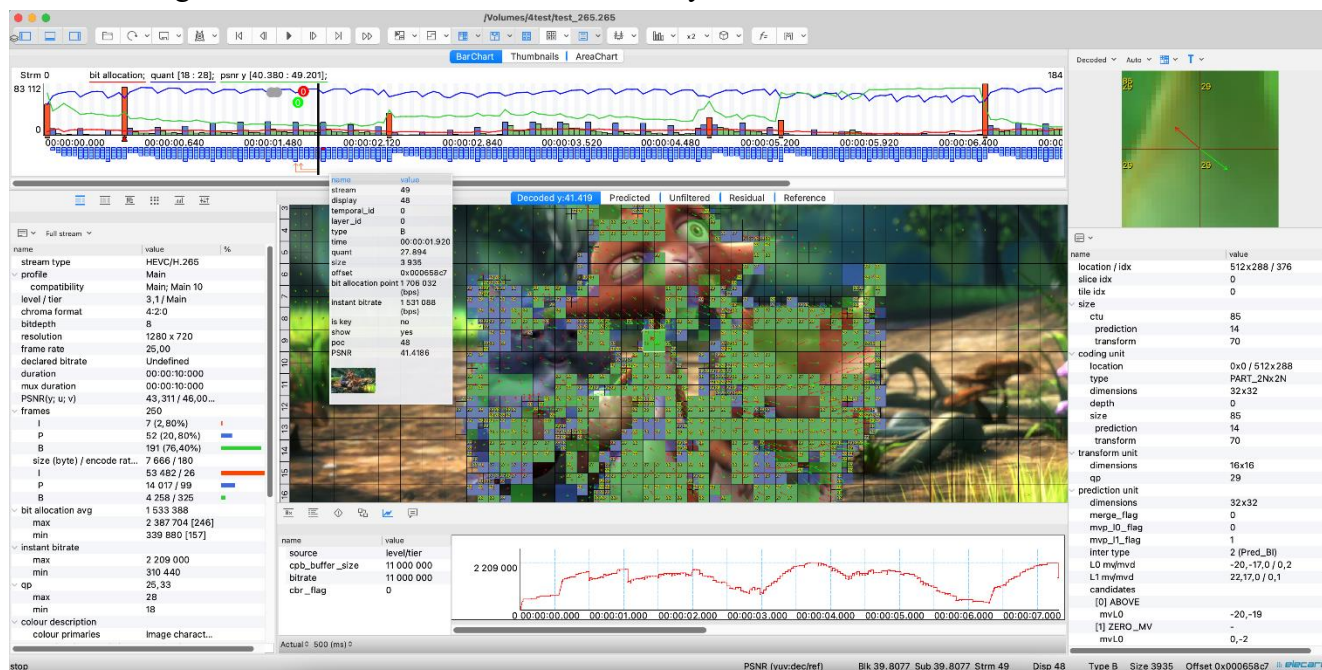


Figure 4. Elecard StreamEye GUI – Main Window

3.2.1 Toolbars

The following table describes the Elecard StreamEye toolbars.

Table 1. Elecard StreamEye GUI – Toolbars

Menu item	Button	Description
Presentation mode		
Left Panel		Hides/shows panels at the left of the application window: <i>Stream Panel</i> , <i>Picture Panel</i> , <i>Headers Panel</i> , <i>Pixels Panel</i> , <i>Statistics Panel</i> , <i>Overlay Panel</i> .
Bottom Panels		Hides/shows panels at the bottom of the application window: <i>Hex Viewer</i> , <i>Stream Viewer</i> , <i>Message Panel</i> , <i>DPB Panel</i> , <i>Buffer Panel</i> , <i>Comments Panel</i> .
Right Panel		Hides/shows panels at the right of the application window: <i>Block Presenter Panel</i> and <i>Block Info Panel</i> .
StreamEye		

About		Shows the <i>About</i> window.
Preferences		Opens the program settings dialog.
Quit StreamEye		Exits the application.
File		
Open file		Opens a new media file for analysis.
Reopen		Reopens the list of recently opened files.
Reopen Last File		Reopens the last opened file.
Save		Saves a stream, picture, headers, index, buffer, block info, YUV data (decoded, predicted, unfiltered, residual), image (decoded, predicted, unfiltered, residual, difference).
Close		Closes the file.
Binding		Enables or disables the <i>Binding</i> mode: <i>Leader</i> — allows sending comments, commands on interface or positioning changes to other applications; <i>Follower</i> — allows receiving comments and commands from other applications.
Open Comments		Opens comments from an XML file.
Save Comments		Saves comments to an XML file.
Open With		Opens the current file with an external application (the application name and its path are set in the <i>Open With List</i> field by opening <i>Options</i> → <i>Main</i> . The player is set by default).
Reference (Full version only)		
Open Reference		Opens a reference stream (raw YUV data).
Reopen Reference		Opens a recently opened reference stream.
Pin		Opens a raw data stream automatically if a media stream is opened.
Pin Type		Always opens a raw data stream if enabled.
Close Reference		Closes the <i>Reference</i> tab.
Reference Configuration		Allows configuring the reference frame.
Navigation		
Go to Begin		Sets the current position to the first frame in the stream.
Step Backward		Sets the current position to the previous frame.
Play		Starts the stream playback.
Step Forward		Sets the current position to the next frame.
Go to End		Sets the current position to the last frame in the stream.
Fast Play		Starts fast playback of the stream.
Step Order		Selects the frame jump mode. The following options are available: <i>Stream</i> , <i>Display</i> , <i>I frame</i> , <i>P Frame</i> , <i>B frame</i> , <i>IP frame</i> , <i>Key frame</i> .
Find...		Opens the <i>Search</i> dialog with search options by stream/display order, time, offset.
Overlays		

Cursor mode		Selects auxiliary info to display with a marker in the <i>Video Out</i> panel: • <i>Partitions</i> — displays the grid and borders of macroblock types. • <i>Tree</i> — displays the hierarchy of division into macroblocks including active CU (for VVC only). • <i>Types</i> — displays a CU type. • <i>MVs</i> — displays motion vectors. • <i>Tooltip</i> — displays a tooltip with block info. • <i>Tooltip config</i> — allows selecting parameters displayed in the tooltip (<i>General</i>, <i>Sizes</i>, <i>Transform</i>, <i>Prediction</i>, <i>Filter</i>). • <i>Metric</i> — calculates a metric for the selected block and displays the metric values in the status bar. • <i>Metric Config</i> — selects a type: PSNR, residual sum of squares, color component (Y, U, V, YUV) and sources for metric calculation (decoded, predicted, unfiltered, reference, after deblocking, after SAO, reference, memory))
Luma/Chroma mode		Performs switching between <i>Luma/Chroma</i> for intra predictions (VVC only).
Slice Boundaries		Displays a subdivision into slices and tiles. (Full version is customizable, the Basic version allows only enabling/disabling the option).
Partitions		Displays prediction and transform units. (Full version is customizable, the Basic version allows only enabling/disabling the option).
Motion Vectors		Displays intra and inter prediction modes, motion vectors, text. (Full version is customizable, the Basic version allows only enabling/disabling the option).
Types		Displays CU types. (Full version is customizable, the Basic version allows only enabling/disabling the option).
Sizes		Displays sizes corresponding to CTU, CU, PU, TU, text. (Full version is customizable, Basic version allows only enabling/disabling the option).
Extended		Displays extended parameters: AVC: <i>Quant</i>, <i>Quant text</i> HEVC: <i>Quant</i>, <i>Quant text</i>, <i>SAO</i>, <i>Merged</i> AV1: <i>Quant</i>, <i>Quant text</i>, <i>Filter CDEF</i>, <i>Filter LR</i> VVC: <i>Quant</i>, <i>Quant text</i>, <i>SAO</i>, <i>ALF</i>, <i>DCT Type</i>* *<i>DCT Type</i> is displayed in the following manner: Lines: - Top line — DCT horizontal - Left line — DCT vertical Colors: - Green — DCT-2 - Red — DCT-7 - Blue — DCT-8 - Yellow rectangle — MTS_SKIP Point in the top left corner: - Blue — LFNST index 1 - Red — LFNST index 2
Filter		(Full version only). Displays the applied filters: • <i>Interlace (frame)</i> — specifies what interlaced frame should be used. • <i>Interlace config</i> — specifies the type of the interlaced frame which should be used. These filters are available only for interlaced frames. • <i>HDR -> SDR</i> — displays the HDR content on an SDR screen (experimental). The filter is available only for AVC, HEVC and 10bit streams.
View		
BarChart		Sets the <i>BarChart</i> presentation parameters: • <i>Presentation Order</i> — changes the display sequence of frames on the <i>BarChart</i>: <i>stream</i> or <i>display</i> order.

		• <i>Scale</i> — sets bar width (1, 2, 4, 8, 16, 32 and 64 pixels). If 32 or 64 pixels are selected, the thumbnails line is displayed. • <i>Vertical Scale</i> — sets bar height: <i>auto</i> or a <i>custom</i> value with saving the last entered value. • <i>Reference</i> — enables the following additional presentation modes for reference frames in the <i>BarChart</i> and <i>Thumbnails</i> navigation controls: ◦ <i>Markers</i> — circles that visualize a reference list for the selected frame. ◦ <i>Lines</i> — arrows to reference parts used at the current frame. ◦ <i>Prediction Parts</i> — shows the percentage of inter prediction L0/L1 (the number of pixels) of the current frame. ◦ <i>Hierarchy</i> — shows the B/P-pyramid visualization. • <i>Bit Allocation Lines</i> — the number of bits processed within -0.5 to +0.5 second at the moment. • <i>Instant Bitrate</i> — the number of bits processed during the last second. • <i>Frame Duration</i> — visualizes the time between the beginning of a frame and the end of that frame. • <i>Temporal ID</i> — visualizes the frame temporal scalability for further analysis. • <i>Sizes</i> — visualizes the distribution of bits in the stream according to the following groups: <i>transform</i>, <i>intra prediction</i>, <i>inter prediction</i>. • <i>Metrics</i> — visualizes the results of metrics calculated from a reference stream. • <i>Quantizers</i> — visualizes the frame mean quantizer value (the blue line). • <i>EPSNR (Full version only)</i> — visualizes the EPSNR metrics values on the <i>BarChart</i> (for MPEG-2 and AVC only). • <i>DPB Occupancy</i> — visualizes the number of frames contained in the DPB decoder; allows estimating the presence of the current frame in the DPB of subsequent frames. • <i>Thumbnails on Tooltip</i> — shows/hides thumbnails on the <i>BarChart</i>. • <i>Graphic Mode</i> — decreases the <i>BarChart</i> brightness to focus attention on graphics. This option is switched on by default.
Scale		Sets the video scaling coefficient: <i>x0.125</i> , <i>x0.25</i> , <i>x0.5</i> , <i>x1</i> , <i>x2</i> , <i>x4</i> , <i>x8</i> .
Layer		Allows to select the active layer when opening multilayer streams. The option allows for selecting one layer or all layers at once.
Chroma format		Selects the displayable color space: <i>YUV</i> , <i>Y</i> , <i>U</i> , <i>V</i> .
Rotate		Sets the video rotation angle: <i>0</i> , <i>90</i> , <i>180</i> , <i>270</i> .
Difference		(Full version only). Enables/disables the difference mode.
Difference type		(Full version only). Enables/disables frame comparison. The combo-boxes <i>Difference Mode</i>, <i>Source 0</i>, <i>Source 1</i> are displayed. The following <i>Difference</i> modes are available: • <i>Compare</i> — a pixel-by-pixel comparison of two frames. If two pixels are matched, the result is 0. Otherwise, the result is equal to the maximum value. • <i>PSNR</i> — the result is PSNR values (from 0 to 99,99) for corresponding CU partitioning blocks, where black color corresponds to 0 value, and white color corresponds to the value of 99,99 and higher. • <i>PSNR Clip</i> — the same as PSNR, but the minimum result value is the minimal PSNR among all blocks in the frame, and maximum is the maximal PSNR, respectively. Black color corresponds to the minimal value, white color corresponds to the maximal value, and blended colors correspond to intermediate values. • <i>Subtraction</i> — pixel-by-pixel subtraction. The result is equal to the difference value. • <i>Temperature</i> — an absolute difference (pixel-by-pixel subtraction); the results are highlighted with black, blue, green and red colors, where black corresponds to zero difference, blue corresponds to a slight difference, green corresponds to a medium difference, and red corresponds to the biggest difference. The following <i>Source 0</i>, <i>Source 1</i> values are available: <i>Decoded</i>, <i>Predicted</i>, <i>Unfiltered</i>, <i>Reference</i>, <i>Memory</i>.

To Memory		(Full version only). Memorizes the selected frame (decoded, predicted, unfiltered, or reference) for use in the <i>Difference</i> analysis. Frames already saved in memory can be removed via the Clear Memory button.
Source 0		(Full version only). Allows selecting the first source for further difference calculation.
Convert		Allows switching between ITU Rec. 601, 709 and 2020 color conversion.
Source 1		(Full version only). Allows selecting the second source for further difference calculation.
Navigation		Allows displaying or activating the <i>BarChart</i> , <i>Thumbnails</i> , <i>AreaChart</i> panels.
Toolbars		Allows selecting the toolbars to be displayed: <i>File</i> , <i>Reference</i> , <i>View</i> , <i>Overlays</i> , <i>Navigation</i> , <i>Help</i> .
Output		Allows displaying or activating the <i>Decoded</i> , <i>Predicted</i> , <i>Unfiltered</i> , <i>AfterDeblocking</i> , <i>AfterSAO</i> , <i>Residual</i> , <i>Reference</i> , <i>Difference</i> , <i>Memory</i> picture output types.
Info		Allows displaying or activating the <i>Stream</i> , <i>Picture</i> , <i>Headers</i> , <i>DPB</i> , <i>Block Info</i> , <i>Block Presenter</i> , <i>Statistics</i> panels.
Tools		Allows displaying or activating the <i>Stream Viewer</i> , <i>HEX Viewer</i> , <i>Pixels</i> , <i>Message</i> , <i>Buffer</i> , <i>Overlay</i> , <i>Comments</i> panels.
Show/Hide		Allows showing/hiding all panels except output pictures.
Show All		Shows all panels.
Hide All		Hides all panels.
Layouts		Allows displaying the selected layouts: view modes from 0 to 3.
Help		
Contents		Opens the current User Guide in .pdf format.
Feedback		Sends the StreamEye feedback via e-mail if it is registered in the system with the following subjects: <i>Submit an Idea</i> , <i>Ask a Question</i> , <i>Report a Bug</i> .

3.2.2 Navigation Controls

To set the stream position to a frame, click the corresponding navigation control: *BarChart*, *Thumbnails*, *AreaChart*. The bit allocation, quant, metrics, instant bitrate, frame duration, sizes, temporal ID, and EPSNR values are displayed on the *AreaChart* as well as on the *BarChart*.

Hover the cursor over a frame, and the frame detailed info will be displayed.

name	value
stream	83
display	83
temporal_id	0
type	B
time	00:00:03.320
quant	40
size	863
offset	0x0001b9d7
bit allocation point	254,760 (bps)
instant bitrate	141,688 (bps)
is key	no
show	yes
poc	83



Figure 5. Elecard StreamEye GUI – Frame Information Window

BarChart:

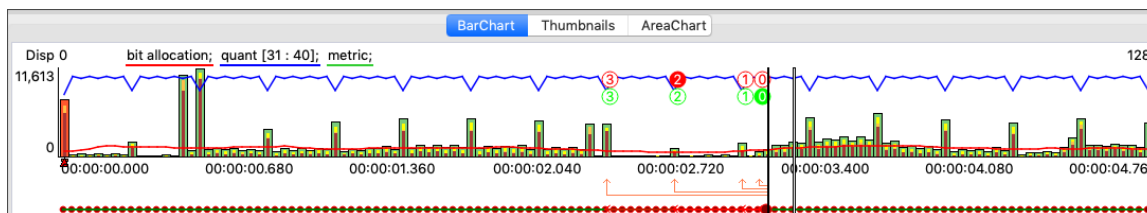


Figure 6. Elecard StreamEye GUI – BarChart Control

The *BarChart* navigation control shows the video stream that is currently open. The bar height illustrates the frame size (in bytes), and the bar color indicates the frame type (red — I, blue — P, green — B).

Thumbnails (Full version only):

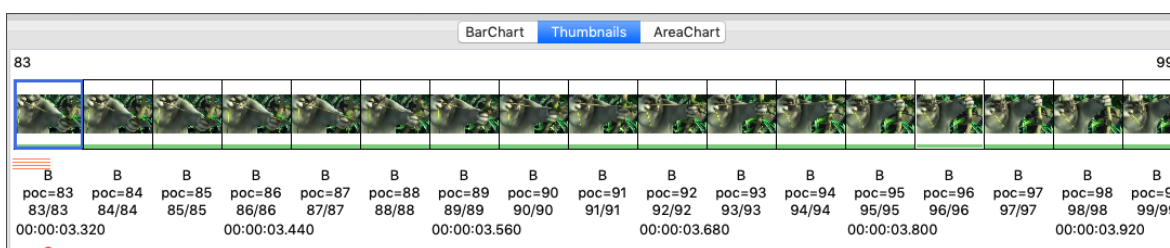


Figure 7. Elecard StreamEye GUI – Thumbnails Control

The *Thumbnails* navigation control shows the stream using frame thumbnails. Colors under thumbnails correspond to the frame type (red — I, blue — P, green — B). The stream/display frame order is indicated with digits below thumbnails.

AreaChart (Full version only):

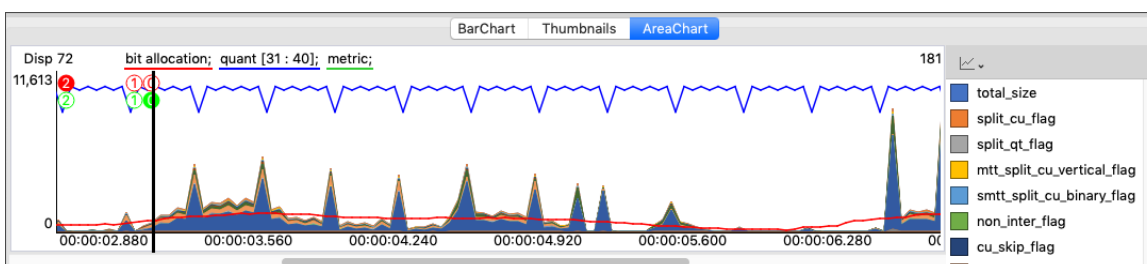


Figure 8. Elecard StreamEye GUI – AreaChart Control

The *AreaChart* navigation control visualizes the distribution of bits in the video stream. The control allows choosing bitstream elements to display. The chart type (area/bars) is configurable via the program settings.

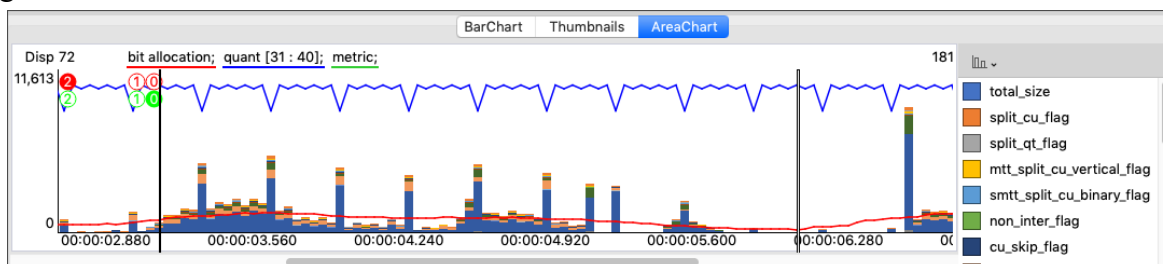


Figure 9. Elecard StreamEye GUI – AreaChart Control (Alternate View)

3.2.3 Comments Panel

The *Comments* panel is designed for team work: share comments on a specific frame or group of frames in a stream, navigation and switching between different instances of the same application or different applications contained in Elecard StreamEye Studio. Comments are created in the *BarChart* panel of the

application by left-clicking (or right-clicking and pressing the shift button) and selecting the required field.

The *Comment* dialog should open. Enter comments in the corresponding field.

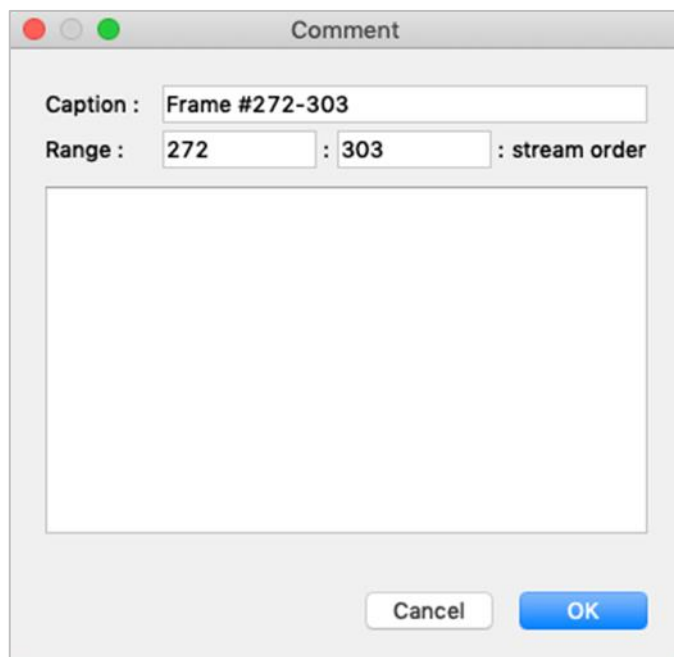


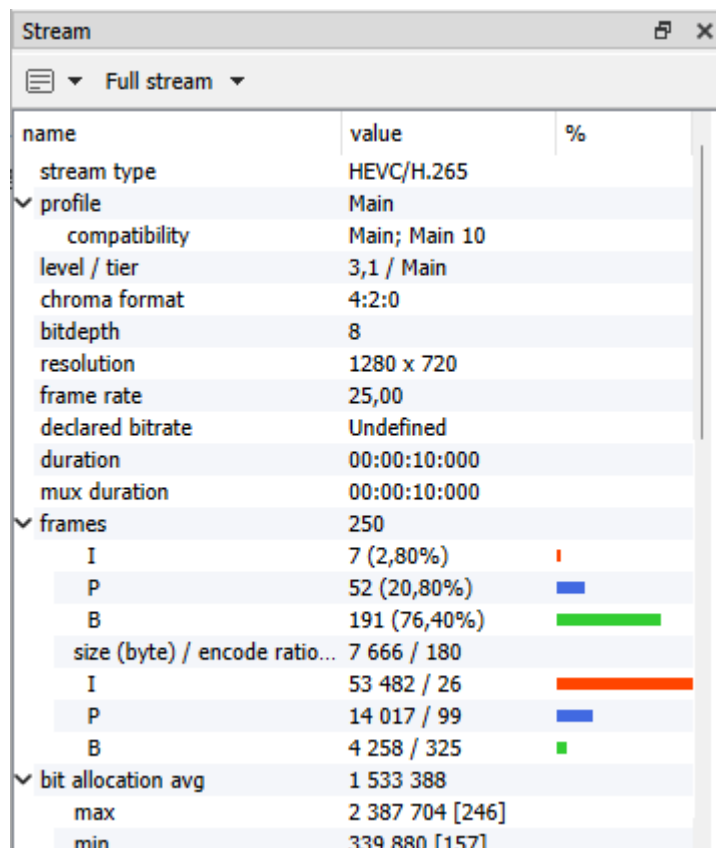
Figure 10. Elecard StreamEye GUI – Comment dialog

The comment parameters can be edited or changed before or after they are saved. To save the comment click *File* → *Save Comments* and select the folder. When the comment is chosen, its position is displayed in the *BarChart* and highlighted green.

To edit, remove or send comments using the *Binding* mode, open the context menu by right-clicking. The *Send* option enables you to pass the selected comment to another application of StreamEye Studio set or another instance of StreamEye. To select an application from which messages should be received or to select a type of messages to be received, go to the *Preferences tab* → *Binding* or see more details in the [Options Dialog](#) section.

3.2.4 Stream Panel

The *Stream* panel displays the stream summary information: profile, compatibility flags, level, tier, chroma format, resolution, frame rate, duration, frame types, average frame sizes, range of quantizer values, bit allocation info, frame average bit distribution in CU.



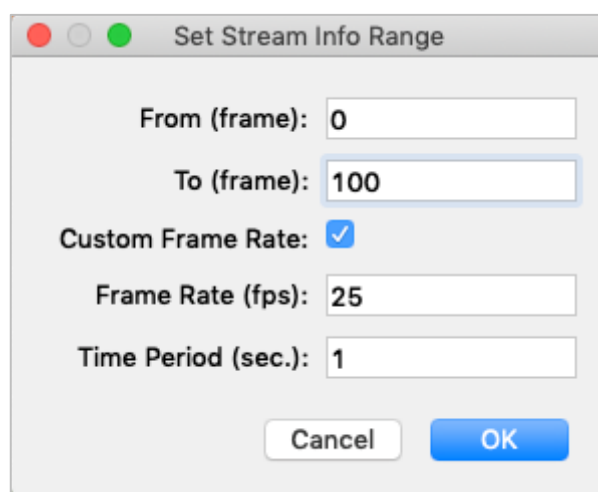
name	value	%
stream type	HEVC/H.265	
profile	Main	
compatibility	Main; Main 10	
level / tier	3,1 / Main	
chroma format	4:2:0	
bitdepth	8	
resolution	1280 x 720	
frame rate	25,00	
declared bitrate	Undefined	
duration	00:00:10:000	
mux duration	00:00:10:000	
frames	250	
I	7 (2,80%)	
P	52 (20,80%)	
B	191 (76,40%)	
size (byte) / encode ratio...	7 666 / 180	
I	53 482 / 26	
P	14 017 / 99	
B	4 258 / 325	
bit allocation avg	1 533 388	
max	2 387 704 [246]	
min	339 880 [157]	

Figure 11. Elecard StreamEye GUI – Stream Panel

The *Group*  selector allows you to select the information displayed.

The *Full stream/Set Range...* switcher allows the user to change the display information mode.

- *Full stream* — shows information about a full stream.
- *Set Range* — allows setting a range of frames within a stream for which summary information should be displayed. The *Custom Frame Rate* checkbox enables specifying a custom fps value. The *Time Period* field value sets the period for which bit allocation and instant bitrate should be recalculated.



Set Stream Info Range

From (frame):

To (frame):

Custom Frame Rate: ☒

Frame Rate (fps):

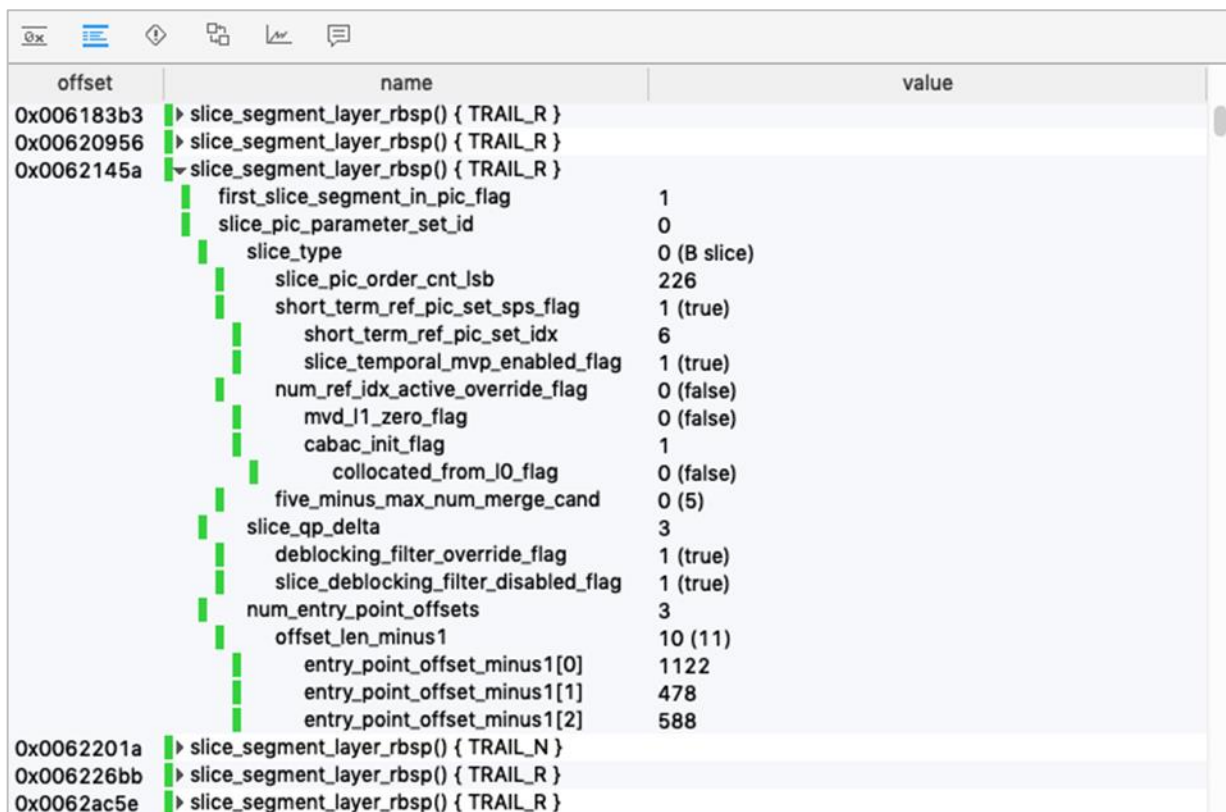
Time Period (sec.):

Figure 12. Elecard StreamEye GUI – Set Stream Info Range Window

3.2.5 Stream Viewer

The *Stream Viewer* displays a structure of headers belonging to the opened stream in the text mode. The

stream internal structures are expandable. Access units are separated with alternative background colors.



offset	name	value
0x006183b3	slice_segment_layer_rbsp() { TRAIL_R }	
0x00620956	slice_segment_layer_rbsp() { TRAIL_R }	
0x0062145a	slice_segment_layer_rbsp() { TRAIL_R }	
	first_slice_segment_in_pic_flag	1
	slice_pic_parameter_set_id	0
	slice_type	0 (B slice)
	slice_pic_order_cnt_lsb	226
	short_term_ref_pic_set_sps_flag	1 (true)
	short_term_ref_pic_set_idx	6
	slice_temporal_mvp_enabled_flag	1 (true)
	num_ref_idx_active_override_flag	0 (false)
	mvd_l1_zero_flag	0 (false)
	cabac_init_flag	1
	collocated_from_10_flag	0 (false)
	five_minus_max_num_merge_cand	0 (5)
	slice_qp_delta	3
	deblocking_filter_override_flag	1 (true)
	slice_deblocking_filter_disabled_flag	1 (true)
	num_entry_point_offsets	3
	offset_len_minus1	10 (11)
	entry_point_offset_minus1[0]	1122
	entry_point_offset_minus1[1]	478
	entry_point_offset_minus1[2]	588
0x0062201a	slice_segment_layer_rbsp() { TRAIL_N }	
0x006226bb	slice_segment_layer_rbsp() { TRAIL_R }	
0x0062ac5e	slice_segment_layer_rbsp() { TRAIL_R }	

Figure 13. Elecard StreamEye GUI – Stream Viewer

The panel content is synchronized with the stream position.

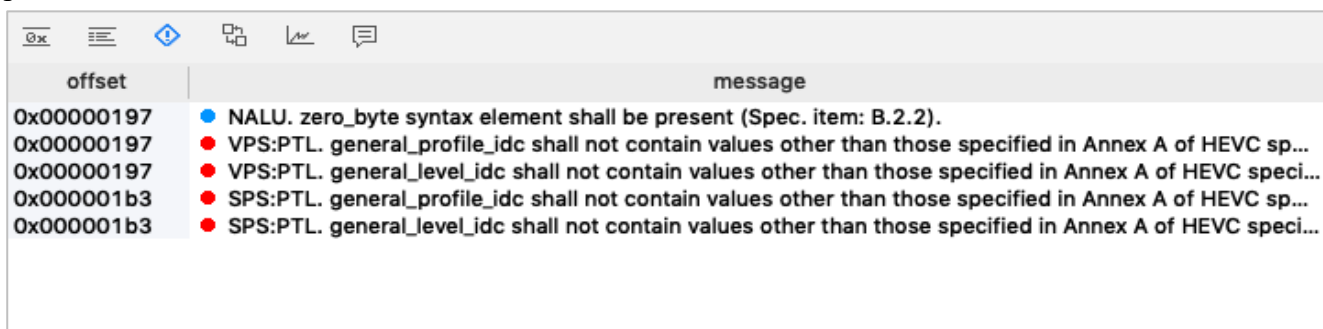
By double-clicking on the offset value, positioning to the related frame is performed in the *BarChart* panel.

3.2.6 Message Panel

(Full version only)

The *Message* panel displays information about compliance verification of the stream against the compression standard. The panel supports display for AVC and HEVC video. For unsupported video formats, nothing is displayed in this panel. Data displayed in the *Message* panel is synchronized with the data displayed in *Hex Viewer* and *Stream Viewer* panels.

By double-clicking on the offset value, positioning to the related frame is performed in the *BarChart* panel.



offset	message
0x00000197	• NALU. zero_byte syntax element shall be present (Spec. item: B.2.2).
0x00000197	• VPS:PTL. general_profile_idc shall not contain values other than those specified in Annex A of HEVC sp...
0x00000197	• VPS:PTL. general_level_idc shall not contain values other than those specified in Annex A of HEVC speci...
0x000001b3	• SPS:PTL. general_profile_idc shall not contain values other than those specified in Annex A of HEVC sp...
0x000001b3	• SPS:PTL. general_level_idc shall not contain values other than those specified in Annex A of HEVC speci...

Figure 14. Elecard StreamEye GUI – Message Panel

3.2.7 Buffer Panel

(Full version only)

To ensure the system operational reliability, the Decoder Video Buffer parameter analysis is performed. The *Buffer* panel displays the results of the analysis (for MPEG-1 and MPEG-2, AVC, HEVC and VVC video formats). For unsupported video formats, nothing is displayed in this panel.

For AVC, HEVC and VVC video formats, the buffer parameters are based on the SEI values. If SEI is not available, the parameters are reconstructed using the stream level/tier (for HEVC video format only).

The following controls are available on the *Buffer* panel:

- *Mode* — sets the chart vertical scaling mode (*Defined* — relative to the *cpb_buffer_size* value; *Actual* — relative to the selected buffer minimum and maximum sizes).
- *Scale* — specifies the chart horizontal scale.
- *Left info panel* — indicates the main parameters that are used for the buffer calculation, and the selected parameter values.

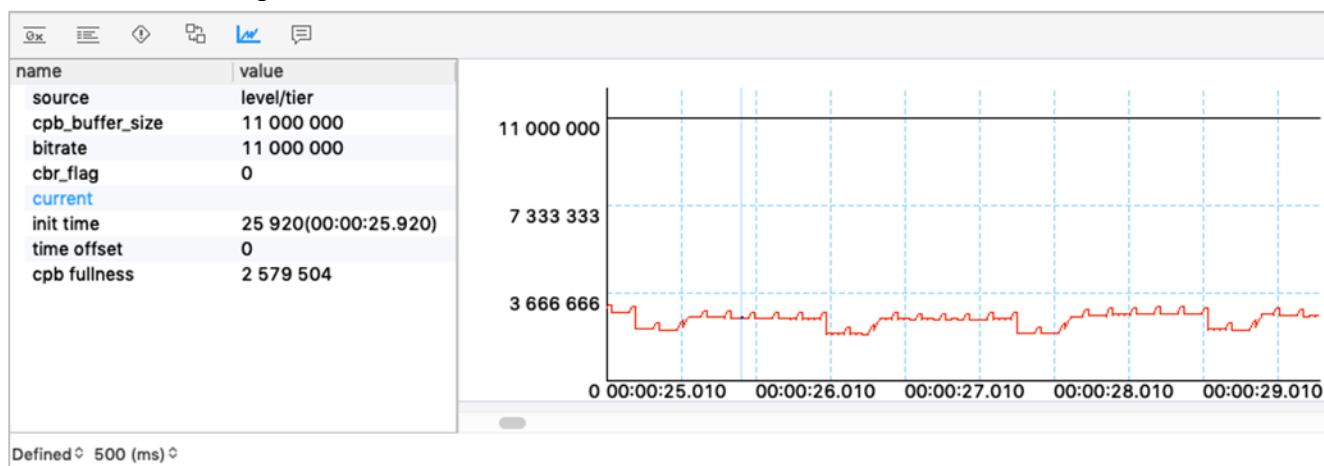


Figure 15. Elecard StreamEye GUI – Buffer Panel

3.2.8 Hex Viewer

The **Hex Viewer** panel contains the following information blocks:

- Offset (from the file beginning) of the line displayed in the next block.
- Part of the file data in the hexadecimal mode.
- Part of the file data in the text mode (ASCII).
- Part of the file data in the binary mode.

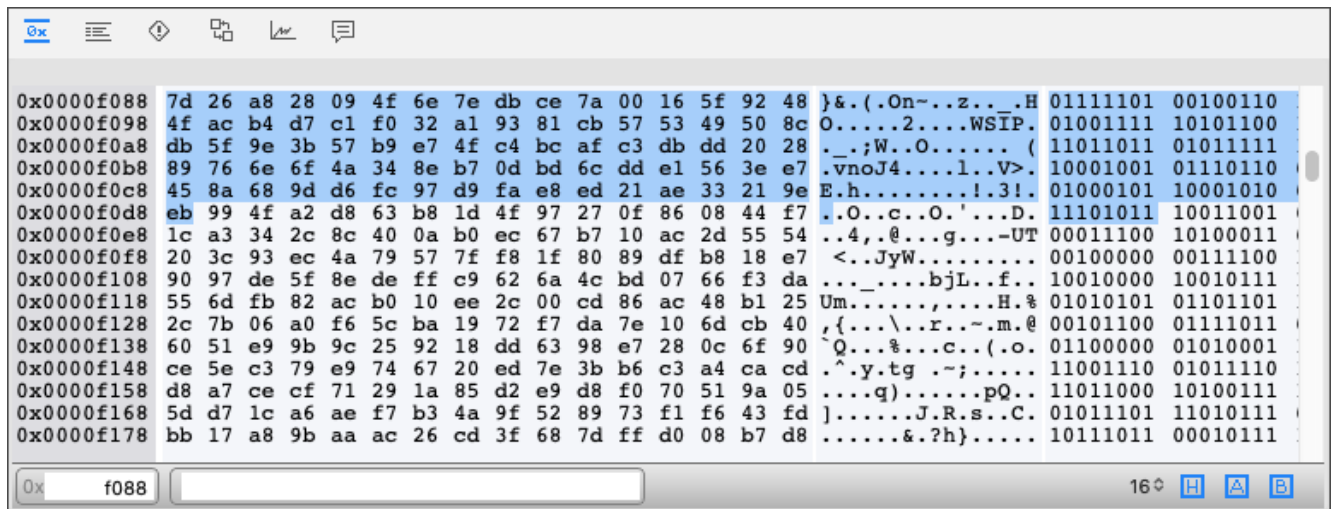


Figure 16. Elecard StreamEye GUI – Hex Viewer

The following table describes the **Hex Viewer** tuning controls.

Table 2. Hex Viewer Controls

Control	Description
	Enables/disables the file data presentation in the hexadecimal mode.
	Enables/disables the file data presentation in the text mode (ASCII).
	Enables/disables the file data presentation in the binary mode.
Columns	Specifies the data amount (the number of bytes) displayed in each line. If the <i>Auto</i> value is set, the length of lines are adjusted to the <i>HEX Viewer</i> panel width.
	Sets the cursor position to the specified offset values.
	Highlights with color those values specified in the corresponding field.

To highlight the data area intended for further copying or saving, hover the cursor, select the required data, and right click it. A drop-down menu will appear. You can copy the data to clipboard or save it to a file:

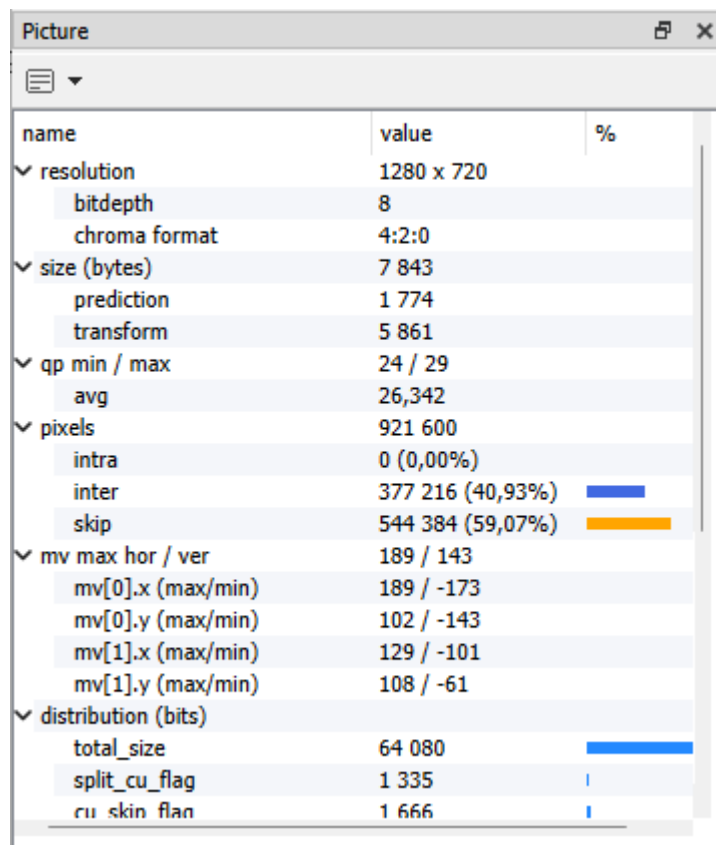
- *Copy to clipboard* — copies the data to clipboard in the text format which depends on the selected HEX/ASCII/BINARY mode.
- *Save to file* — saves the selected binary data to a file.

The panel content is synchronized with the stream position.

3.2.9 Picture Panel

The *Picture* panel displays the frame summary information: CU, PU, TU size (in bits), max/min QP, pixel distribution into encoded type (intra, inter, and skip), motion vectors' range, bit distribution in CU.

The *Group* selector allows you to select the information displayed.



name	value	%
✓ resolution	1280 x 720	
bitdepth	8	
chroma format	4:2:0	
✓ size (bytes)	7 843	
prediction	1 774	
transform	5 861	
✓ qp min / max	24 / 29	
avg	26,342	
✓ pixels	921 600	
intra	0 (0,00%)	
inter	377 216 (40,93%)	
skip	544 384 (59,07%)	
✓ mv max hor / ver	189 / 143	
mv[0].x (max/min)	189 / -173	
mv[0].y (max/min)	102 / -143	
mv[1].x (max/min)	129 / -101	
mv[1].y (max/min)	108 / -61	
✓ distribution (bits)		
total_size	64 080	
split_cu_flag	1 335	
cu_skin_flag	1 666	

Figure 17. Elecard StreamEye GUI – Picture Panel

3.2.10 Headers

(Full version only)

The *Headers* panel displays bitstream headers that are used for displaying and decoding the selected frame (as in the Standard documentation).

Right click the required line to copy or to copy with a child. The *Copy* function allows copying the selected line. The *Copy with child* function allows copying the child record when copying the parent record.

Hover the cursor over a parameter, and a pop-up window showing information on the corresponding standard will appear.

To filter header parameters, enter the required word in the search field.

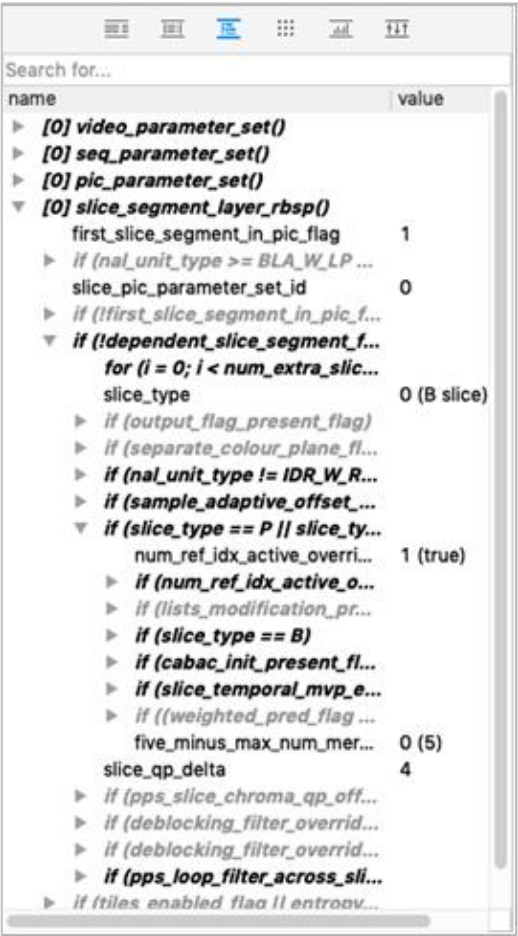


Figure 18. Elecard StreamEye GUI – Headers Panel

3.2.11 DPB Panel

(Full version only)

The *DPB* panel indicates the Decoded Picture Buffer information for the selected frame. The DPB panel supports displaying information for AVC, HEVC, VVC, VP9, and AV1 video formats. For unsupported video formats, nothing is displayed in this panel.

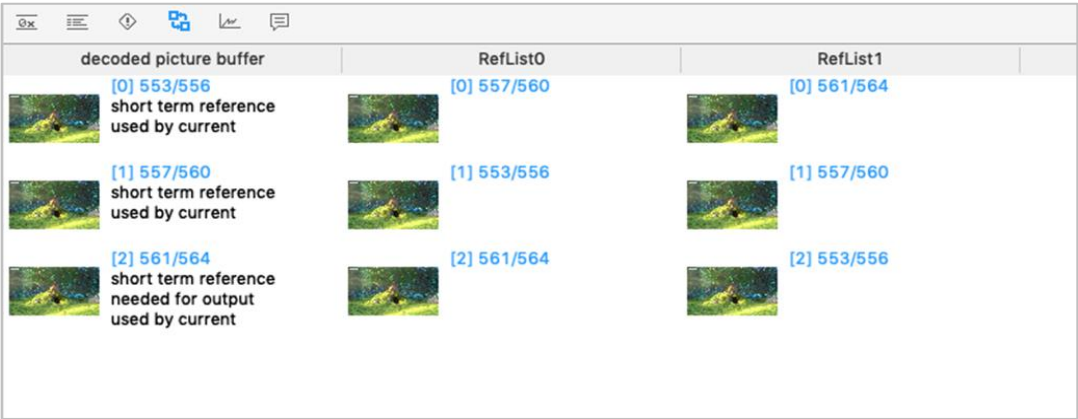
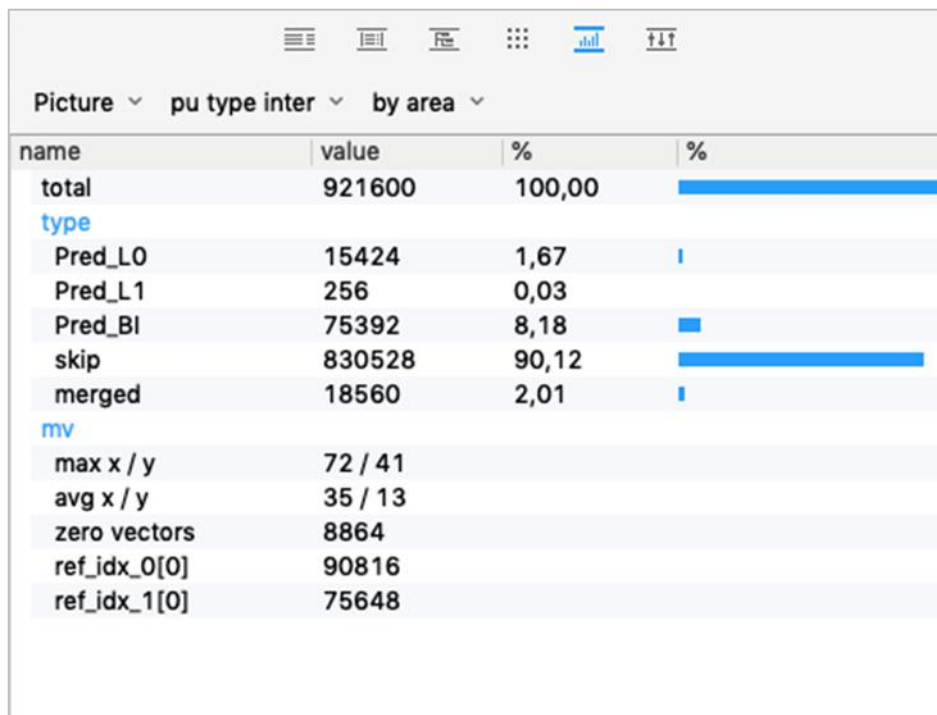


Figure 19. Elecard StreamEye GUI – DPB Panel

3.2.12 Statistics Panel

(Full version only)

The *Statistics* panel displays statistics on streams and selected frames. Statistics details depend on a media file format. The common structure covered by the statistics includes the distance between *I frames*, *P Frames*, *B frames* and *Key frames*.



name	value	%	%
total	921600	100,00	
type			
Pred_L0	15424	1,67	
Pred_L1	256	0,03	
Pred_BI	75392	8,18	
skip	830528	90,12	
merged	18560	2,01	
mv			
max x / y	72 / 41		
avg x / y	35 / 13		
zero vectors	8864		
ref_idx_0[0]	90816		
ref_idx_1[0]	75648		

Figure 20. Elecard StreamEye GUI – Statistics Panel

Statistics data can be shown either for a stream or for a picture and either by count, or by area:

- *Stream* — displays statistics for the whole stream by:
 - *Structure* — the most common and average values are displayed.
 - *Prediction* — indicates the percentage of various prediction types in the stream.
- *Picture* — depending on the stream format, the statistics is displayed for each format of the selected frame:
 - **HEVC format:**
 - *CU info* — counts partition types and dimensions applied for Coded Units.
 - *PU type intra* — counts Prediction Intra Types (luma/chroma) applied for Prediction Units.
 - *PU type inter* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for Prediction Units.
 - *TU info* — counts dimensions, zero sizes applied for Transform Units.
 - *SAO info* — shows SAO types applied for luma and chroma, using merge_up_flag and merge_left_flag.
 - **AVC format:**
 - *MB info* — counts types & dimensions applied macroblocks.
 - *Prediction type intra* — counts Prediction Intra Types (luma/chroma) applied for macroblocks.
 - *Prediction type inter* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for macroblocks.
 - *Transform info* — counts dimensions, the number of zero sizes applied for macroblocks.
 - **MPEG-1/2 format:**

- *MB info* — counts mb types applied for macroblocks.
- **VP9 format:**
 - *Block info* — counts dimensions applied for blocks.
 - *Intra modes* — counts prediction intra types applied for blocks
 - *Inter modes* — counts prediction inter types and vectors' data such as max x/y, avg x/y, zero vectors count, LAST_FRAME/GOLDEN_FRAME/ALTREF_FRAME applied for blocks.
- **AV1 format:**
 - *Block info* — counts dimensions applied for blocks.
 - *Intra modes* — counts prediction intra types applied for blocks
 - *Inter modes* — counts prediction inter types and vectors' data such as max x/y, avg x/y, zero vectors count, NEARESTMV/NEARMV/GLOBALMV/NEWMV/NEAREST_NEARESTMV/NEAR_NEARMV/NEAREST_NEWMV/NEW_NEARESTMV/NEAR_NEWMV/NEW_NEARMV/GLOBAL_GLOBALMV/NEW_NEWMV applied for blocks.
- **VVC format:**
 - *CU info* — counts partition types and dimensions applied for Coded Units.
 - *PU type intra* — counts Prediction Intra Types (luma/chroma) applied for Prediction Units.
 - *PU type inter* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for Prediction Units.
 - *TU info* — counts dimensions, zero sizes applied for Transform Units.
 - *TU type* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for Transform Units.
- **AVS3 format:**
 - *CU info* — counts dimensions applied for Coded Units.
 - *PU type intra* — counts Prediction Intra Types (luma/chroma) applied for Prediction Units.
 - *PU type inter* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for Prediction Units.
 - *Prediction* — counts numbers of predictions in the frame for various CU sizes. The mean prediction error (MPE) is the residual mean absolute value. The value describes the prediction mean (for pixel) deviation from the decoded picture that characterizes the prediction quality.
 - *TU info* — counts dimensions applied for Transform Units.
 - *TU type* — counts Prediction Inter Types (luma/chroma) and vectors' data such as max x/y, avg x/y, zero vectors count, ref_idx_0/refidx_1 lists applied for Transform Units.

3.2.13 Video Out Panels

The panel displays decoded information at various steps (decoded, predicted, unfiltered, residual and after deblocking, after SAO (for VVC only)), corresponding reference frame, and selected difference.

For decoded, predicted, and unfiltered tabs, indication (in the tab name) of PSNR calculated with reference for Average, Y, U, V is available (adjusted via the program settings).

The scroll wheel button allows the video scaling from 12,5% to 800%.

Click the selected CTU to lock its position.

Click and drag to select an area.

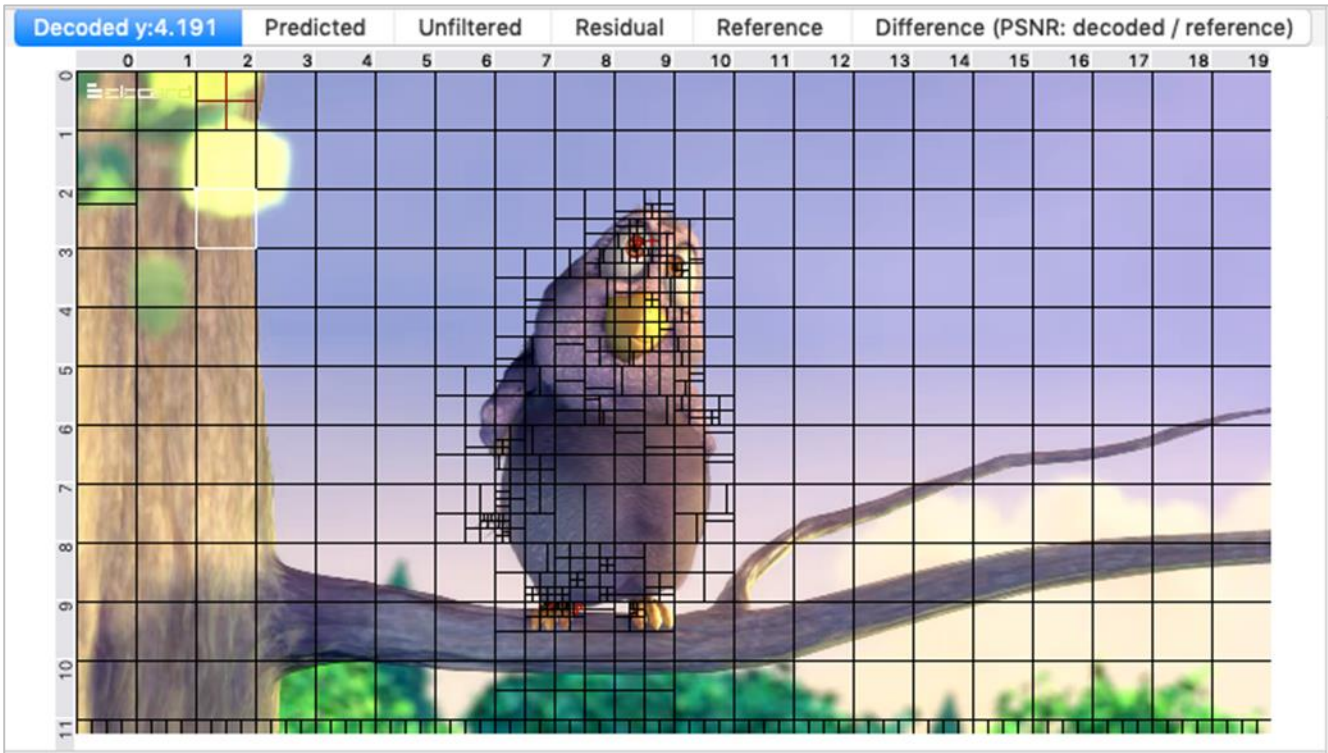


Figure 21. Elecard StreamEye GUI – Video Out Panel – Decoded Tab

Point to the selected Prediction subblock/Block in the difference (PSNR clip/decoded) tab, to display in the tooltip the PSNR values for Y, U, V components.

3.2.14 Overlay Panel

(Full version only)

The panel is used for *Overlay* configuration and allows setting the required parameters.

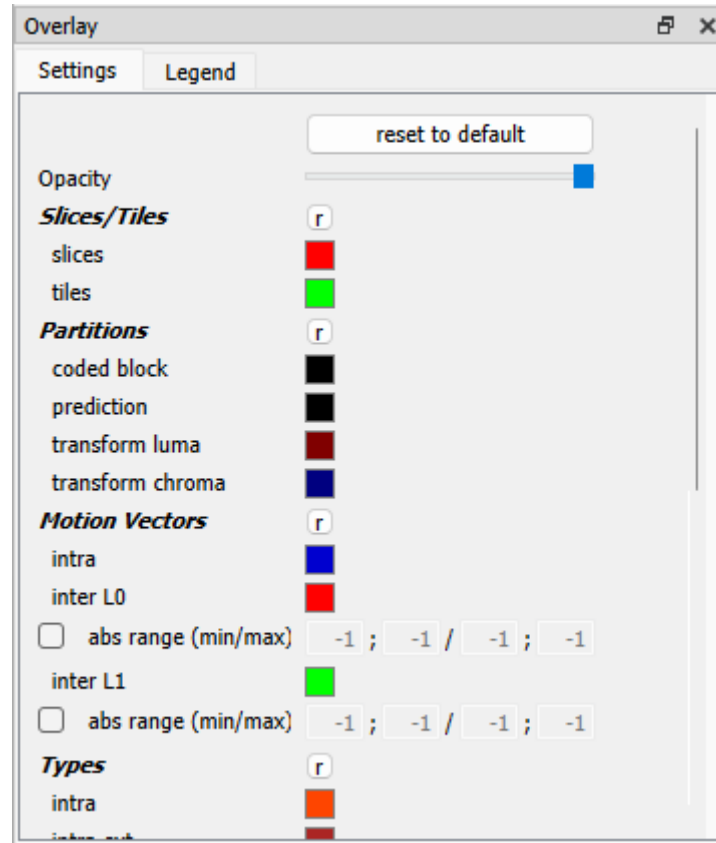


Figure 22. Elecard StreamEye GUI – Overlay Settings Panel

- *Opacity* — transparency of the *Overlay*.
- *Slices/Tiles* — allows selecting a color for slices and tiles display.
- *Partitions* — allows selecting a color for partitions display.
- *Motion Vectors* — allows selecting a color for motion vectors display. The range -1; -1 / -1; -1 is set by default and disregarded. If the range is enabled and the values are set, the areas matching the stated values only are displayed.
- *Types* — allows selecting a color for intra/inter one direction/inter two direction/inter direct (AVC) display.
- *Sizes* — allows selecting a color for block sizes display.
- *Quant* — allows selecting a color for quantizer display.
- *Difference PSNR* — a range showing a difference in PSNR blocks. Blocks not matching the specified range are not displayed.
- *Text* — allows setting a color and shadow of extended parameters text displaying.
- *Overlay Legend Panel* — displays symbols for the current overlay settings.

3.2.15 Block Presenter Panel

The *Block Presenter* panel provides the visualization of the selected block for smooth navigation and detailed information on CU parameters.

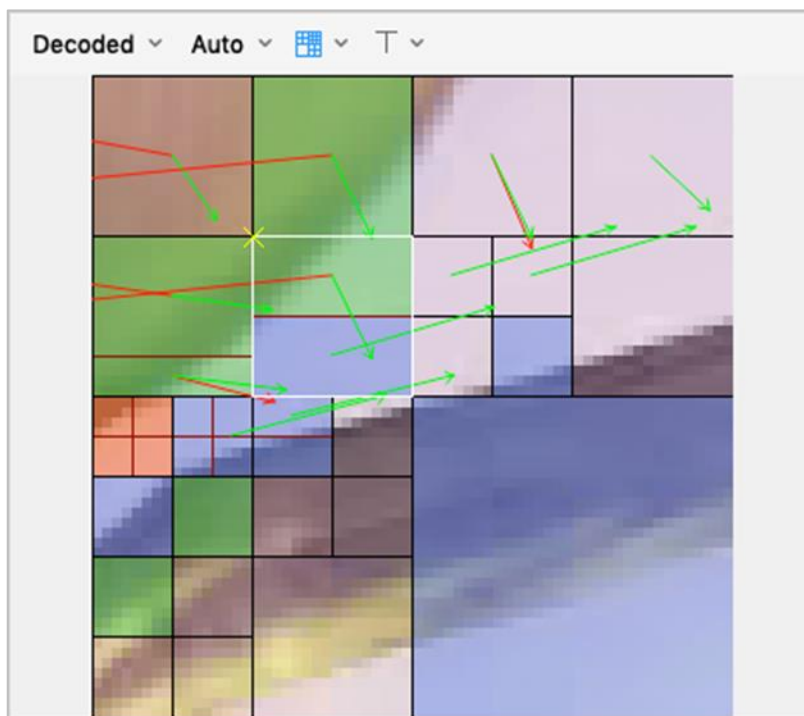


Figure 23. Elecard StreamEye GUI – Block Presenter Panel

The following table describes the *Block Presenter* panel controls.

Table 3. Block Presenter Panel Controls

Control	Description
Type	Allows selecting a processing stage of the stream: • <i>Decoded</i> • <i>Predicted</i> • <i>Unfiltered</i> • <i>Residual</i>
Scale	Scaling by: <i>x2</i> , <i>x4</i> , <i>x8</i> , <i>x16</i> , <i>Auto</i> . If the <i>Auto</i> mode is selected, the frame size is automatically adjusted to the control size.
	Shows, hides, and configures the overlay parameters: • <i>Partitions</i> • <i>Motion Vectors</i> • <i>Types</i> • <i>Background</i>
T	Shows, hides, and displays values for the selected blocks: • <i>Predictions</i> • <i>Sizes</i> • <i>Quantizers</i>

3.2.16 Block Info Panel

The panel displays block information (Macroblock/Coding Tree Unit/Super Block) depending on the opened stream type and includes: location, slice index, tile index, sizes, coded unit, predicted unit, transform unit info and information for SAO (HEVC, VVC), CDEF, LR (AV1), ALF (VVC).

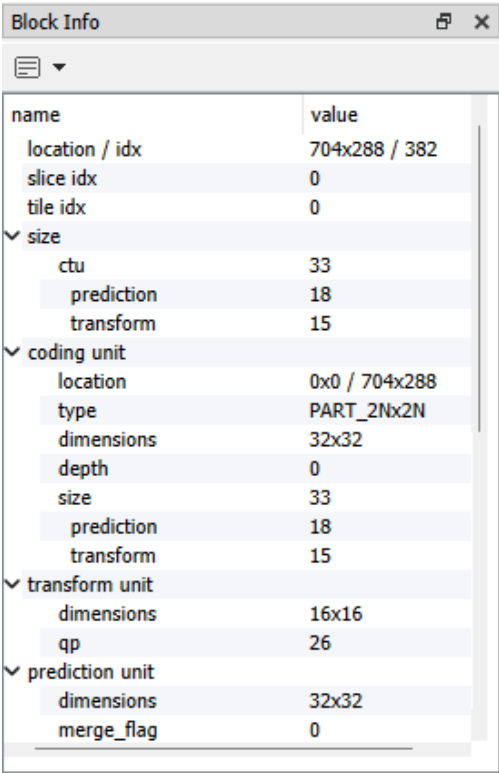


Figure 24. Elecard StreamEye GUI – Block Info Panel

Table 4. Block Info Panel Controls

Control	Description
	Shows, hides and displays values for the selected blocks: • <i>General</i> • <i>Sizes</i> • <i>Transform</i> • <i>Prediction</i> • <i>Filter</i>

3.2.17 Pixels Panel

(Full version only)

The *Pixels* panel displays pixel YUV values for different decoding steps (decoded, predicted, unfiltered and after decoding, after SAO (for VVC only), coefficient values (residual, transform, dequantized), reference (if opened), and difference (if selected and exists).

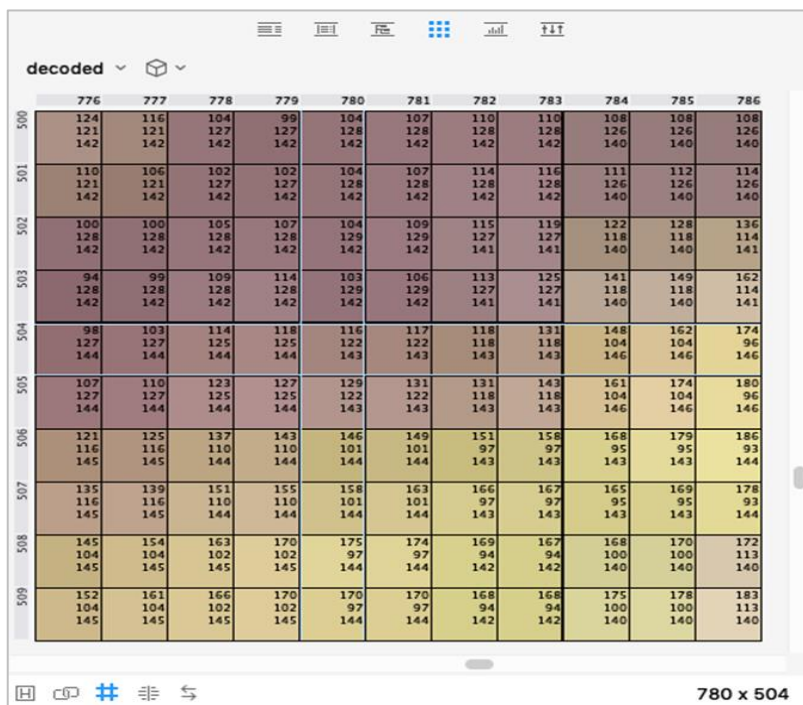


Figure 25. Elecard StreamEye GUI – Pixels Panel

The following table describes the *Pixels* panel controls.

Table 5. Pixels Panel Controls

Control	Description
	Allows displaying data in two windows for comparison.
After Deblocking ▾	Displays decoded information for selected frame at various steps (decoded, predicted, unfiltered, residual, difference, transform, dequantized). For VVC, the After deblocking and after SAO modes are also available.
	Selects YUV components for visualization. The background color corresponds to the pixel color.
	Snaps the reference point to the selected block on the video output.
FF	Displays data in decimal/hexadecimal mode.
	Displays the partitions grid.
	Synchronizes the visualization mode according to the active tab of the <i>Video Out</i> panel.

To select the visualization area, left-click it. To copy data, right-click it, and the drop-down menu will appear: copy subblock, copy block.

3.2.18 Options Dialog

To configure file settings, its visualization, performance, binding and saving modes, open the *Options* dialog by opening the StreamEye menu command and selecting the *Preferences* tab.

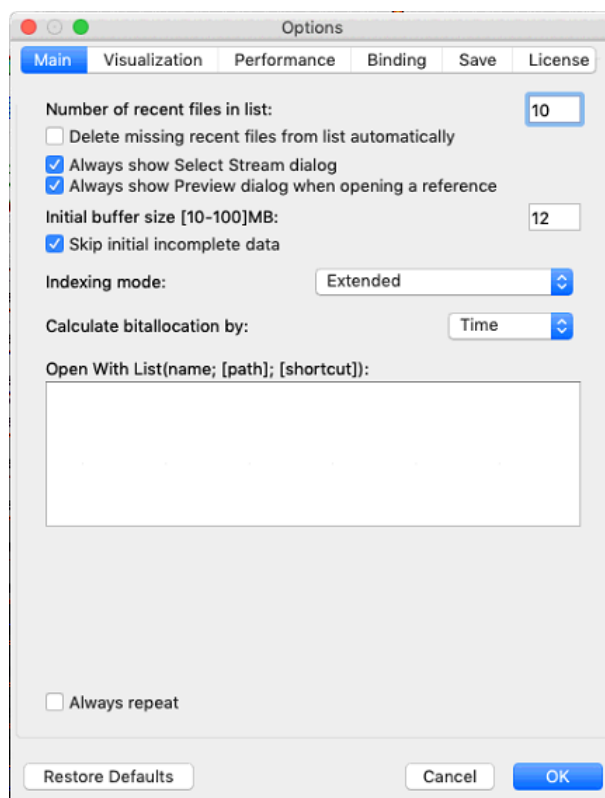


Figure 26. Elecard StreamEye GUI – Options Dialog

The following table describes the *Main* tab in the *Options* dialog.

Table 6. Elecard StreamEye – Options dialog – Main tab

Control	Function
Number of recent files in list	Shows the number of recent files being saved in a list.
Delete missing recent files from list automatically	Allows deleting missing files from a recent list automatically when opening a file from the list.
Always show Select Stream dialog	Enables the display of the <i>Select Stream</i> dialog on a permanent basis.
Always show Preview dialog when opening a reference	Enables/disables the display of the <i>Preview</i> dialog when opening a stream. If the option is disabled, the stream is opened with the last used type, offset, and increment parameters.
Initial buffer size 10-100 MB	Allows specifying a parser initialization buffer.
Skip initial incomplete data	Allows skipping data that precedes the first basic headers: • AVC: SPS, PPS; • HEVC: VPS, SPS, PPS; • MPEG-2: sequence_header, picture_header
Indexing mode	Starts the indexing mode and allows selecting its type: • <i>Extended</i> — performs parsing (defines number, type, time, and basic size per each frame) and indexing (detailed statistics for frames and thumbnails are collected during indexing). • <i>Base</i> — performs parsing only.
Calculate bitallocation by:	• <i>Time</i> — calculates bitallocation based on the times from a container (if available). • <i>FrameRate</i> — calculates bitallocation based on the frame rate.

Open with list	Allows opening the file currently opened in StreamEye with another application (see details below).
Always repeat	Allows looping a media file, i.e. when the media file reaches its end, it will immediately start playing again from the beginning.

The *Options* dialog allows opening a file with an external application. To open the file currently opened in StreamEye with another application, specify its name and path in the *Open With List* field according to parameters specified there. The parameters path and shortcut are optional. If the path parameter is not specified, default application or player supporting a file format and installed in the system is opened. Each string specified in the *Open With List* field corresponds to an individual item in the menu *File* → *Open with* that allows easy switching between applications or players after configuring the field.

Note: The **Restore Defaults** button allows you to reset set options to their original values. Functionality of this button is not applicable to the *License* tab.

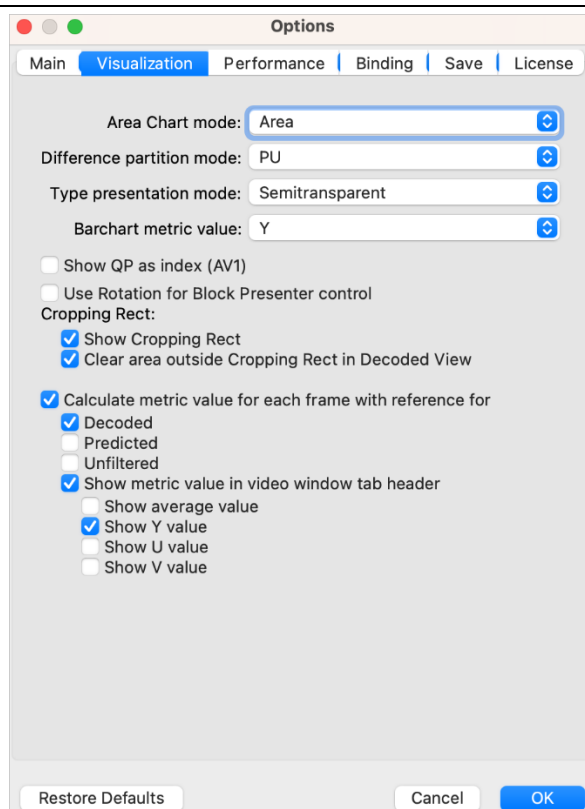


Figure 27. Elecard StreamEye GUI – Options Dialog – Visualization Tab

The following table describes the *Visualization* tab in the *Options* dialog.

Table 7. Elecard StreamEye – Options dialog – Visualization tab

Control	Function
AreaChart mode	Defines the type of the <i>AreaChart</i> presentation: • <i>Area</i> — solid line • <i>Bar</i> — columns
Difference partition mode	Allows selecting a partition mode when calculating <i>PSNR</i> and <i>PSNR Clip</i> between prediction subblocks and blocks.
Type presentation mode	Allows choosing between types to be displayed in the overlay: <i>semitransparent</i> , <i>rectangle</i> .
Barchart metric value	Allows selecting a component to be displayed on the <i>BarChart</i> : <i>Y</i> , <i>U</i> , <i>V</i> , <i>YUV</i> .

Show QP as Index (AV1)	Allows showing quantizer parameter values as index (for AV1)
Use Rotation for Block Presenter control	Allows the rotation of the selected block in the <i>Block Presenter</i> panel when the entire frame is rotated.
Cropping Rect	Adjusts the display of the cropping area in the <i>Video Out</i> panel: <i>Show cropping rect</i> — display cropping applied to the video as a red line; <i>Clear area outside Cropping Rect in decoded view</i> — disables the display of frame information in the cropping area.
Calculate metric value for each frame with reference for:	Allows metrics calculation for the selected parameters: <i>Decoded</i> <i>Predicted</i> <i>Unfiltered</i> <i>Show metric value in video window tab header</i> — allows selecting metric values to be displayed in tab headers: <i>Show average value</i> <i>Show Y value</i> <i>Show U value</i> <i>Show V value</i>
Text size	Allows customizing the font size of the interface: <i>Auto, 8, 10, 12, 14</i> .

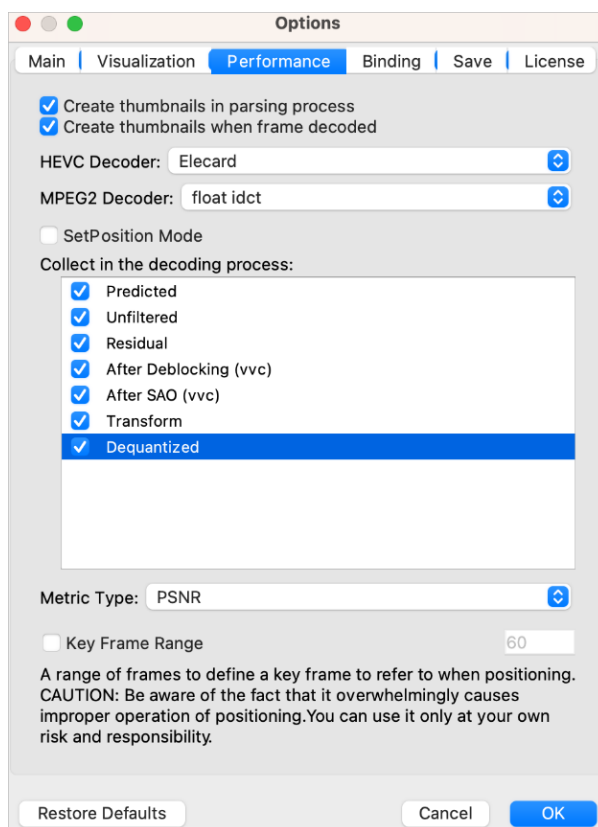


Figure 28. Elecard StreamEye GUI – Options Dialog – Performance Tab

The following table describes the *Performance* tab in the *Options* dialog.

Table 8. Elecard StreamEye – Options dialog – Performance tab

Control	Function
Create thumbnails in parsing process	Enables or disables the creation of thumbnails in the parsing process (saves CPU's power).
Create thumbnails when frame decoded	Enables or disables the creation of thumbnails when a frame is decoded (saves CPU's power).

HEVC Decoder	Allows selecting a decoder for HEVC decoding: <i>reference</i> or <i>Elecard</i> .
MPEG2 Decoder	Allows selecting a decoder for MPEG2 decoding: <i>float idct</i> or <i>int idct</i> .
SetPosition Mode	Enables faster frame positioning. The enabled mode slows down the playback start speed.
Collect in the decoding process	Allows specifying the amount of data for parsing: <i>Predicted</i> , <i>Unfiltered</i> , <i>Residual</i> , <i>After Deblocking (vvc)</i> , <i>After SAO (vvc)</i> , <i>Transform</i> , <i>Dequantized</i> . The amount of data affects the application performance speed.
Metric Type	Allows selecting metric types to be calculated and displayed: <i>APSNR</i> , <i>Delta</i> , <i>MSAD</i> , <i>MSE</i> , <i>NQI</i> , <i>PSNR</i> , <i>SSIM</i> , <i>VIF</i> , <i>VMAF</i> , <i>VMAF (Phone)</i> , <i>VQM</i> .
Key Frame Range	Allows specifying a range of frames to define a key frame to refer to when positioning. Note: Be aware of the fact that it overwhelmingly causes improper operation of positioning. You can use it only at your own risk and responsibility.

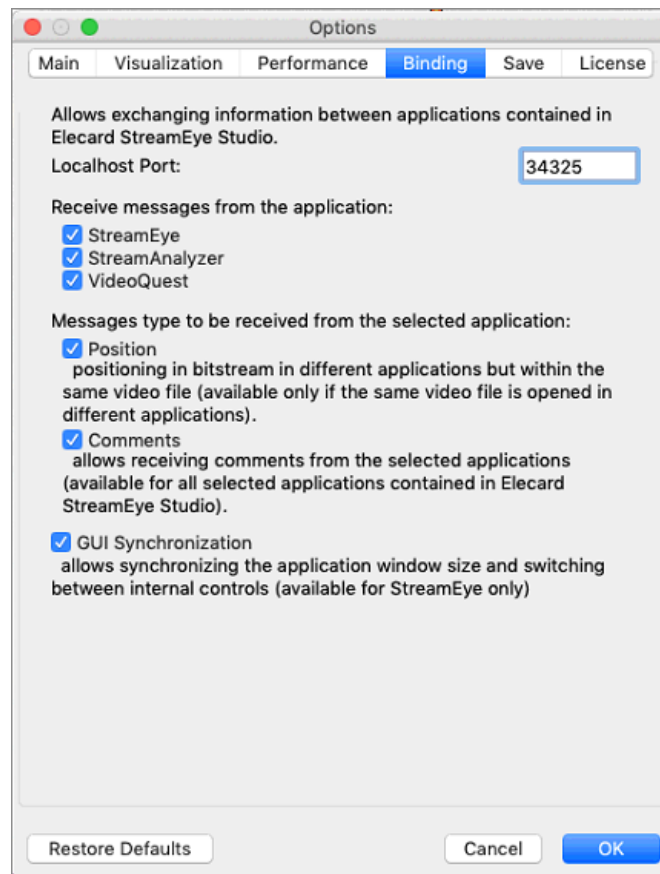


Figure 29. Elecard StreamEye GUI – Options Dialog – Binding Tab

The *Binding* tab contains settings that allow configuration of the interaction between Elecard StreamEye Studio applications. In the *Binding* mode, exchanging information between applications is provided via a specified port. By activating checkboxes, you can select which applications should send messages and set messages type that should be communicated. Additionally, the synchronization of applications window size and switching between controls can be enabled.

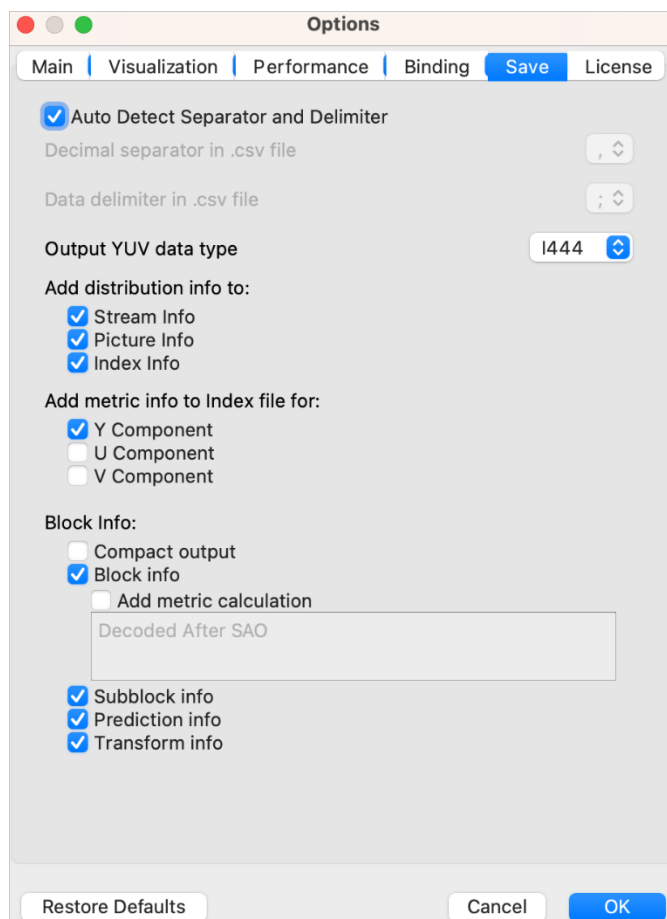


Figure 30. Elecard StreamEye GUI – Options Dialog – Save Tab

The following table describes the *Save* tab in the *Options* dialog.

Table 9. Elecard StreamEye – Options dialog – Save tab

Control	Function
Auto Detect Separator and Delimiter	Allows automatically detecting decimal separators and data delimiters in .csv files
Decimal separator in .csv file	Allows selecting a decimal comma or a decimal dot to separate an integer part from a fractional one of a real number.
Data delimiter in .csv file	Allows selecting a comma or a semicolon to specify boundaries in the data stream.
Output YUV data type	Allows saving YUV data in one of the following formats: <i>I444</i> , <i>I422</i> , <i>IYUV</i> , <i>NV12</i> , <i>NV21</i> , <i>P420</i> , <i>P444</i> , <i>RGB24</i> , <i>RGB32</i> , <i>UYVY</i> (<i>Y422</i> , <i>UYNV</i> , <i>HDYC</i>), <i>V210</i> , <i>V400</i> , <i>V444</i> , <i>Y42B</i> , <i>YUY2</i> (<i>YUNV</i> , <i>V422</i> , <i>YUYV</i>), <i>YV12</i> , <i>YV16</i> , <i>YVYU</i> .
Add distribution info to	Allows adding distribution info when saving a file: <i>Stream Info</i> , <i>Picture Info</i> , <i>Index Info</i> .
Add metric info to Index file for	Allows adding metric info for the selected parameters when saving: <i>Y Component</i> , <i>U Component</i> , <i>V Component</i> .

Block info	Saving parameters of block information:
	• <i>Compact output</i> — combines information in one line under the block parameters. • <i>Block info</i> — sets output parameters. - <i>Add metric calculation</i> — allows calculating metrics for the selected block. • <i>Subblock info</i> — saves CU information. • <i>Prediction info</i> — saves PU information. • <i>Transform info</i> — saves TU information.

3.2.19 Preview Dialog

(Full version only)

The section gives an overview of the *Preview* dialog. The window is displayed when the *Reference* configuration menu item is selected or the  button is clicked. If the *Always show this dialog when opening a reference* checkbox is set, the *Preview* dialog pops up automatically when opening a reference stream.

If the reference stream has not been opened, the  button is not active. The parameter depends on the stream resolution that is currently open and cannot be changed.

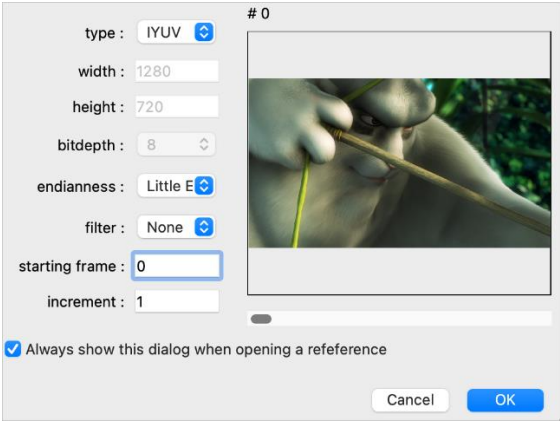


Figure 31. Elecard StreamEye GUI – Preview Dialog

The following table describes the *Preview* dialog controls.

Table 10. Elecard StreamEye – Preview dialog

Control	Function
Type	Specifies the type of the YUV data during the analyzing process.
Width	Displays the width value of a frame which cannot be changed.
Height	Displays the height value of a frame which cannot be changed.
Bitdepth	Allows the user to specify values for some metric types. For several types, the value is not editable.
Endianness	Allows the user to select the order of bytes in the upsampler: <i>Little Endian</i> or <i>Big Endian</i> .
Filter	Selects a preprocessing filter.
Starting frame	Allows specifying a starting frame (from the beginning of a reference stream).
Increment	Specifies an increment value, i.e. a coefficient used for calculation of a successive frame number intended for further analysis. 1 is set by default.

3.2.20 Hot Keys

For tabs, functions and modes switching press hot keys described hereto.

- Ctrl + O – opens a **media file**.
- Ctrl + R – opens the **recently opened file**.

The following hot keys combinations are used for **Navigation**:

- Ctrl + Shift + Left – starts the **Go to beginning** function
- Alt + Left – goes to the **Previous** frame
- Ctrl + Space – starts the **Play** function
- Alt + Right – goes to the **Next** frame
- Ctrl + Shift + Right – starts the **Go to end** function

To change **View** configuration of panel controls, use the following hot keys:

- F2 – shows/hides **all controls**
- F3 – shows **all** (controls)
- F4 – hides **all** (controls)
- F5, F6, F7, F8 – change **layout configuration**
- F9 – shows **top** panel controls (shows/hides all controls being located at the top of the application).
- F10 – shows **bottom** panel controls (shows/hides all controls being located at the bottom of the application).
- F11 – shows **left** panel controls (shows/hides all controls being located on the left of the application).
- F12 – shows **right** panel controls (shows/hides all controls being located on the right of the application).

For configuring **overlays**, use the following hot key combinations:

- ALT+1 – shows/hides the **slice boundaries** overlay
- ALT+2 – shows/hides the **partition** overlay
- ALT+3 – shows/hides the **motion vectors** overlay
- ALT+4 – shows/hides the **types** overlay
- ALT+5 – shows/hides the **sizes** overlay
- ALT+6 – shows/hides the **extended** overlay

The following hot keys are used for switching **Video Out Panels**:

- CTRL+1 – activates the **decoded** panel
- CTRL+2 – activates the **predicted** panel
- CTRL+3 – activates the **unfiltered** panel
- CTRL+4 – activates the **residual** panel
- CTRL+5 – activates the **reference** panel
- CTRL+6 – activates the **difference** panel that displays the selected difference (for instance PSNR clip or decoded)

4. Command Line Tool

(Full version only)

StreamEye Command Line tool is designed for insight into MPEG-1/2, H.264/AVC, H.265/HEVC, VP9, AV1, and AVS3 video compression standards and ever fast quality and compliance check.

StreamEye Command Line tool allows automatic analysis of video arrays. You can customize a sequence of commands to perform your specific tasks and automate your routine tasks.

Launch of Command Line Tool

StreamEye Command Line tool is located in the following path: /Users/bin/SEConsole/SeyeConsole4. To streamline automatic analysis of several files the following rules are introduced. They reassign input and output parameters for configuration files.

Get to the command prompt by clicking Start->Run and type 'cmd'. Then enter the command to start the application first and proceed with the command line options. In general, the syntax is as follows:

```
./streameye-cli config.xml /input test.mp4 /reference test.yuv /output test_out
```

Note: You should not relocate the <product> console without all the other applications of the product contained in the installation pack, otherwise the console application will not start.

Notes on the command line rules:

1. Parameters with default value are not required and can be omitted;
2. Rules for input and output file names:
 - parameter --input [path] in the command line overwrites the value of the tag <input/> in the configuration file;
 - parameter --output [path] in the command line overwrites the value of the tag <output/> in the configuration file;
 - if <output/> tag is missing (or not specified), output file names are formed from input file name plus suffix (if the latter is specified);
 - if output file for a tag/section is not specified, the output file name for the tag/section is formed from the name specified in the <output/> tag plus suffix depending on the name of the tag/section, e.g. ".stream.csv", ".picture.csv", etc.

4.1 Configuration File

Sample Config.xml file contains all the commands performed by the application and reflects overall operation results:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="..." />
  <output separator=";(default)/," delimiter=".(default)/," file="..." />
  <reference type="IYUV(default)/[*Upsampler Type]"
  filter="None(default)/FieldGen/Median/Sobel" offset="0(default)"
  increment="1(default)" file="..." />
  <range span="all(default)/start - stop" order="display(default)/time"
  indexmode="base/extended(default)" />
  <info>
    <stream distribution="on(default)/off" file="..." />
    <picture distribution="on(default)/off" file="..." />
  </info>
</stream>
```

```

<data
type="decoded(default)/predicted/unfiltered/residual/reference/difference
" format="iyuv(default)/rgb32/wide" file="..."/>
<message level="all(default)/message/warning/error" file="..."/>
<header level="struct(default)/parameter" file="...">
    <condition parameter="name" comparison="full/substring(default)">
        <range value="min-max"/>
        <equal value="value"/>
        <less value="min"/>
        <more value="max"/>
        <not value="value"/>
    </condition>
</header>
<index distribution="on(default)/off" order="stream(default)/display"
file="..."/>
<metrics type="psnr(default)/delta/msad/mse/ssim/vqm/vmaf/nqi/apsnr/vmaf
phone"/>
source="decoded(default)/predicted/unfiltered/residual/reference/differen
ce" file="..."/>
<difference type="compare/psnr/psnr clip/subtraction/temperature"
source0="decoded(default)/predicted/unfiltered/residual/reference"
source1="reference(default)/decoded/predicted/unfiltered/residual"
file="..."/>
<buffer file="..."/>
<block compact="false(default)/true"
disable="block|subblock|prediction|transform" file="..."/>
<dpb description="on(default)/off" file="..."/>
<statistics mode="all(default)/stream/picture"
type="all(default)/[*Statistics Type]"
prediction="all(default)/sizes/mpe" component="all(default)/luma/chroma"
by="all(default)/count/area" file="..."/>
</info>
</stream>

```

* Upsampler types that can be used in a configuration file: I444 / I422 / IYUV / NV12 / P420 / P444 / RGB24 / RGB32 / UYVY / V210 / V400 / Y42B / YUY2 / YV12 / YV16 / YVYU / NV21 / V444.

4.2 List of Commands

List of commands contains information on the commands and all parameters which may be used within a command, and their description.

Table 11. List of Commands

Command	Description
<input/> file	The tag is used to configure the input file; Sets the input file path.
<output/> separator delimiter file	The tag is used to configure the input file; Allows selecting decimal comma or decimal dot to separate an integer part from a fractional one of a real number. Allows selecting comma or semicolon to specify boundaries in data stream. Sets the output file path.
<reference/> type filter offset	The tag is used to configure the reference file; Defines the format of the raw data. YV12 (is set by default) / *Upsampler type are used now. More formats to be supported in the next updates); Enables filtering in a frame (None (by default)/Field Generation/Median/Sobel); Offset from the beginning up to the specified frame in a reference stream. It is expressed in frames. "0" is set by default;

Command	Description
increment file	Coefficient used for calculation of a successive frame number intended for further analysis. “1” is set by default; Sets the path to the reference file.
<range/> span order index mode	The tag is used to define the range of analyzed frames; Defines the range of frames (all (by default) /start - stop), 'start' — the number of the first analyzed frame, 'stop' — the number of the last analyzed frame; Specifies the order of frames. Display is set by default; Starts indexing mode (extended (by default) / base).
<info/>	The tag is used to define the type/-s of information for output.
<stream/> distribution file	The tag is used to output general information about the analyzed stream; Contains information on data distribution by type of coding units (on (by default) / off); Sets the path for the output stream information.
<picture/> distribution file	The tag is used to output picture information; Contains information on data distribution by type of coding units (on (by default) / off); Sets the path for the output picture information.
<data/> type format file	The tag is used to save output decoder information on any decoding step; Specifies the type of data for saving (decoded (by default) / predicted / unfiltered / residual / raw data / difference); Defines the format for saving data (IYUV (by default)); Sets the output file path where the data will be saved.
<message/> level file	The tag is used to output messages that appear during analysis; Defines the level of the saved messages (all (by default) / message / warning / error); Sets the output file path where the messages will be saved.
<header/> level file	The tag is used to define the output headers information; Defines the type of header output information (struct (by default) / parameter), 'struct' — full information on the header will be saved, 'parameter' — only information on the specified parameter(s) from the headers will be saved; Sets the path to the output file with the headers information.
<condition/> parameter comparison	The tag is used to specify the required header parameters. If this section does not exist, all headers will be dumped; Specifies the header parameter name; Specifies the type of comparison check for the parameter name (full / substring (by default)).
<range/> value	The tag is used to specify the range of parameter values to be searched; Defines the range for searching parameter values in the format “min-max&rdquo”.
<equal/> value	The tag is used to set the search parameter value equal to the specified value; Sets the parameter value for comparison.
<less/> value	The tag is used to set the search parameter value less than the specified value; Sets the parameter value for comparison.
<more/> value	The tag is used to set the search parameter value exceeding the specified value; Sets the parameter value for comparison.
<not/> value	The tag is used to set the search parameter value NOT equal to the specified value; Sets the parameter value for comparison.
<index/> distribution order file	The tag is used to output information about frames (offset, frame type, frame size, average quantizer, etc.); Contains information on data distribution by type of coding units (on (by default) / off); Shows the order or sequence of index data that are saved in stream/display; Sets the output file path where information on frames will be saved.
<metrics/>	The tag is used to output information on quality metrics;

Command	Description
type	Specifies the metric type: PSNR (by default) SSIM DELTA MSAD MSE VQM NQI APSNR VMAF phone;
source	Defines the type of the source for metrics calculation (decoded (by default) / predicted / unfiltered);
file	Sets the output file path where information on metrics will be saved.
<difference/>	The tag is used to output information on difference metrics
type	Defines the type of difference (compare/subtraction/temperature). This is the required parameter for this tag. Its default value should be set. If the value is not set, this tag will be skipped;
source0	Specifies the source type for difference calculation (decoded (by default) / predicted / unfiltered / residual/ reference);
source1	Specifies the source type for difference calculation (reference (by default) / decoded / predicted / unfiltered / residual);
file	Sets the output file path where information on difference will be saved.
<buffer/>	The tag is used to save Decoder video buffer parameters;
file	Sets the path to the output file with the buffer information
<block/>	The tag is used to define information about each encoding block;
compact	Data presentation type (false (by default)/true), false — each parameter of one block is presented in one line; true — combines a block data and presents as a column;
disable	Excludes data from the report (block and/or subblock and/or prediction and/or transform);
file	Sets the output file path where information on difference will be saved.
<dpb/>	The tag is used to save information about decoded picture buffer for each frame; description on (by default)/off is used to display or not the additional information about the frame.
description	Enables/disables adding of frame description from dpb panel;
file	Sets the output file path where information on difference will be saved.
<statistics/>	The tag is used to get statistics about streams and frames (see more details in the Statistics Panel section).
mode	Allows selecting a mode (stream/pictures) for getting statistics.
type	Shows statistics for the whole stream by structure or prediction.
prediction	Counts prediction types depending on the file format and components.
component	Allows selecting components for counting predictions.
by	Allows selecting the way of displaying statistics (by count/by area).
file	Sets the output file path where information on difference will be saved.

These examples show how to perform some of the typical tasks from the command-line:

1. [Saving the general information on the stream and on each picture;](#)
2. [Retrieving information on the number of IDR_W_RADL slices and offsets;](#)
3. [Searching for all slices with the slice type value equal to "2";](#)
4. [Saving YUV data on the prediction step;](#)
5. [Saving the block info data of the picture;](#)
6. [Retrieving information on frame sizes and PSNR values for the first 10 frames](#)

1. Saving the general information on the stream and on each picture

config.xml:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="C:\Media\hevc\test_12_0_sei.h265"/>
  <output file="C:\Media\hevc\out\"/>
  <info>
    <stream/>
    <picture/>
  </info>
</stream>
```

```
</info>
</stream>
```

Output files:

test_12_0_sei.stream.csv:

name;	value;
stream type;	HEVC/H.265;
profile;	Main;
compatibility;	Main; Main 10;
level / tier;	Undefined / Main;
chroma format;	4:2:0;
resolution;	832 x 480;
frame rate;	50.00;
bitrate;	1 024 000 000;

...

test_12_0_sei.picture.csv:

Picture #0;	
name;	value;
resolution;	416 x 240;
size (bytes);	
cu;	21 476;
prediction;	520;
transform;	19 264;
qp;	
min / max;	23 / 23;
pixels;	
count;	99 840;
intra;	99 840 (100.00%);
distribution (bits);	
total;	174 352 (100.00%);
split_cu_flag;	297 (0.17%);
cu_skip_flag;	0 (0.00%);
prediction;	
intra_prediction;	16 653 (9.55%);
prediction;	
intra_prediction;	16 653 (9.55%);
...	
Picture #1:	
name;	value;
resolution;	416 x 240;
size (bytes);	
cu;	978;
prediction;	352;
transform;	511;
...	

test_12_0_sei.picture.csv:

Picture #0;	
-------------	--

name;	value;
resolution;	416 x 240;
size (bytes);	
cu;	21 476;
prediction;	520;
transform;	19 264;
qp;	
min / max;	23 / 23;
pixels;	
count;	99 840;
intra;	99 840 (100.00%);
distribution (bits);	
total;	174 352 (100.00%);
split_cu_flag;	297 (0.17%);
cu_skip_flag;	0 (0.00%);
prediction;	
intra_prediction;	16 653 (9.55%);
prediction;	
intra_prediction;	16 653 (9.55%);
...	
Picture #1:	
name;	value;
resolution;	416 x 240;
size (bytes);	
cu;	978;
prediction;	352;
transform;	511;
...	

2. Retrieving information on the number of IDR_W_RADL slices and offsets

config.xml:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="C:\Media\hevc\test_12_0_sei.h265"/>
  <output file="C:\Media\hevc\out\"/>
  <info>
    <header level="parameter">
```



```

        <condition parameter="IDR_W_RADL" comparison="substring"/>
    </header>
</info>
</stream>

```

Output file:

test_12_0_sei.header.csv:

offset;	name;
0x000000AA;	slice_segment_layer() { IDR_W_RADL };
0x00025F47;	slice_segment_layer() { IDR_W_RADL };

3. Searching for all slices with the slice_type value equal to "2"

config.xml:

```

<?xml version="1.0"?>
<stream version="1">
    <input file="C:\Media\hevc\test_12_0_sei.h265"/>
    <output file="C:\Media\hevc\test_12_0_sei"/>
    <info>
        <header>
            <condition parameter="slice_type">
                <equal value="2"/>
            </condition>
        </header>
    </info>
</stream>

```

Output file:

test_12_0_sei.header.csv:

offset;	name;	value
0x0000009C;	slice_segment_layer() { IDR_W_RADL };	
	first_slice_segment_in_pic_flag;	1;
	no_output_of_prior_pics_flag;	0;
	slice_pic_parameter_set_id;	0;
	slice_type;	2;
	slice_sao_luma_flag;	1;
	slice_sao_chroma_flag;	1;
	slice_qp_delta;	6;
	slice_loop_filter_across_slices_enabled_flag;	1;
0x0000D21A;	slice_segment_layer() { CRA_NUT };	
	first_slice_segment_in_pic_flag;	1;
	no_output_of_prior_pics_flag;	0;
	slice_pic_parameter_set_id;	0;

	slice_type;	2;
	slice_pic_order_cnt_lsb;	32;
	short_term_ref_pic_set_sps_flag;	0;
	short_term_ref_pic_set(12);	
	if(stRpsIdx != 0);	
...		

4. Saving YUV data on the prediction step

config.xml:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="C:\Media\hevc\test_12_0_sei.h265"/>
  <output file="C:\Media\hevc\out\"/>
  <info>
    <data type="predicted"/>
  </info>
</stream>
```

Output file:

test_12_0_sei.data.416x240.yuv

5. Saving the block info data of the picture

config.xml:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="C:\Media\hevc\test_12_0_sei.h265"/>
  <output file="C:\Media\hevc\out\"/>
  <info>
    <block/>
  </info>
</stream>
```

Output file:

test_12_0_sei.block10.csv

name;	value;
lcu location;	0x0;
lcu slice\tile idx;	0\0;
lcu size total\prediction\transform;	17\17\0;
cu type\location\dimension\depth;	PART_2Nx2N\0x0\64x64\0;
cu size total\prediction\transform;	17\17\0;
tu dimensions\qp;	64x64\35;
Pu inter dimension\merge flag\mvp 10 flag\mvp 11 flag;	64x64\0\0\0;
pu inter type\dimension;	2 (Pred_BI)\64x64;

pu L0 mv\mvd;	-1, 0, 1 / -1, 0;
pu L1 mv\mvd;	-1, 1, 0 / -1, 1;
pu mv candidates;	*[0]\mvL0(0,0)\[1]\mvL0(0,0)*0]\mvL1(0,0)\1]\mvL1(0,0);
lcu location;	64x0;
lcu slice\tile idx;	0\0;
...	

6. Retrieving information on frame sizes and PSNR values for the first 10 frames

config.xml:

```
<?xml version="1.0"?>
<stream version="1">
  <input file="C:\Media\hevc\test_12_0_sei.h265"/>
  <output file="C:\Media\hevc\out\"/>
  <reference file="C:\Media\raw\BasketballDrill_832x480_50.yuv"/>
  <range span="0-10"/>
  <info>
    <metrics type="psnr"/>
    <index distribution="off" order="display"/>
  </info>
</stream>
```

Output file:

test_12_0_sei.index.csv:

stream;	display;	offset;	poc;	type;	time;	quant;	psnr y;
0;	0;	0x00000000;	0;	I;	00:00:00:000;	32;	36.065069;
1;	8;	0x000042B3;	8;	B;	00:00:00:160;	33;	36.065069;
2;	4;	0x00005790;	4;	B;	00:00:00:160;	34;	36.065069;
...							
10;	12;	0x00008343;	12;	B;	00:00:00:240;	34;	34.808986;

4.3 Features of Command Line Tool

- Analysis the whole array of video files automatically;
- Addressing specific and complex challenges with a few lines of commands;
- Retrieving submitted results as a text file in CSV format for easy viewing in Excel;
- Customizing a sequence of commands to perform your specific tasks;
- Accelerating and automating your regular tasks;
- Getting access to most of StreamEye functionality and resources through StreamEye Console.

5. Comparison Table

The following table describes differences between Elecard StreamEye (Full version) and Elecard StreamEye Basic.

Table 12. Elecard StreamEye – Comparison Table

		StreamEye (Full version)	StreamEye Basic
Features	Functionality		
Video Formats	<i>Video Formats that are supported</i>	AVC, MPEG-1/2, HEVC, VVC, VP9, AV1, AVS3	
Reference / Raw file support	<i>Comparison of the encoded file with the reference / raw file</i>	Allows more accurate compression quality estimation as compared with no-reference estimation. Available metric types: Delta, MSAD, MSE, PSNR, APSNR, SSIM, VQM, VMAF, VMAF phone, NQI, VIF at various steps: decoded, predicted, unfiltered, residual.	Not available
Binding mode	<i>Comparison of similar blocks or frames of two streams encoded with different settings from the same raw file</i>	Allows synchronization of the selected frame and block position with similar frame and block of a stream opened via another StreamEye instance running in the system.	Not available
Stream Presentation/Charts	<i>Statistics visual representation. Easy navigation throughout the stream using several stream presentation types</i>	BarChart The bar height illustrates the frame size in bytes and the bar color indicates the frame type. Presentation order, Scale, Bitallocation lines, reference markers, sizes, metrics, quantizers, EPSNR, DPB Occupancy parameters are shown. Tooltip with thumbnails, frame position in the stream, type, timestamp, quantizer, offset, picture order count, size is displayed for each frame.	All parameters of the BarChart are available except for EPSNR (Estimated Pick to Signal Noise Ratio).
		AreaChart The AreaChart navigation control shows bit distribution in the stream. The control allows choosing bitstream elements to display: skipped macroblocks, macroblock and prediction types, transform data, flags, quarter delta. Tooltip is added.	Not available Not available
		Thumbnails The Thumbnails navigation control shows the video stream using the frame thumbnails. The thumbnails are underlined with color indicating the frame type. The stream/display frame order is specified with digits below thumbnails. Tooltip is added.	
Info panels	<i>Detailed information on each and any component of the stream</i>	Stream The Stream panel displays the stream summary information: profile, compatibility flags, level, tier, chroma	All parameters of the Stream panel are available except for bit distribution in the coded elements.

		format, resolution, frame rate, duration, frame types, average frame sizes; range of quantizer values; bit allocation info; frame average bit distribution in CU.	
		Picture The panel displays the frame summary information: CU, PU, TU size (in bits); max/min QP; pixel distribution into encoded type (intra, inter, and skip); motion vectors range; bit distribution in CU.	All parameters of the Picture panel are available except for bit distribution in the coded elements.
		Block info The panel displays Macroblock /Superblock/Coding Tree Unit/TCU information: location, slice index, tile index; sizes; coded unit, predicted unit, transform unit info.	
		Headers The panel displays the bitstream headers that are used for the selected frame decoding and display. Header parameter values along with indication of their compliance to the standards are shown.	Not available
		DPB info The panel shows the Decoded Picture Buffer information for the selected frame. The DPB panel supports display for AVC and HEVC video.	Not available
		Block Presenter The panel provides block visualization for smooth navigation and detailed information on CU parameters. Shows processing stage of a stream, overlay parameters: partitions, motion vectors, types, background; values for the selected block: predictions, sizes, quantizes.	Not available
		Statistics The panel displays statistics on streams or selected frames and can be shown either by count, or by area: distance between I frames, P Frames, B frames and Key frames; macroblock types and dimensions, prediction and transform info and additional information depending on the stream type.	Not available
Tools	<i>Helpful instruments addressing specific areas of investigation during bitstream analysis</i>	Stream Viewer The stream viewer displays a structure of headers belonging to the opened stream in the text mode. The stream internal structures are expandable.	
		HEX Viewer The viewer displays a stream in the HEX mode. The display is synchronized with the stream position.	
		Pixels Viewer The viewer displays pixel YUV values for different decoding steps (decoded, predicted, unfiltered), coefficient values (residual, transform, dequantized), reference (if opened) and difference (if selected and exists).	Not available

		Message Viewer The viewer displays messages on compression standard compliance for the stream. The Message viewer supports display for AVC and HEVC video.	Not available
		Buffer Analyzer To ensure the system operational reliability, the Decoder Video Buffer parameter analysis is performed. The Buffer viewer displays results of the analysis (for AVC and HEVC video formats). The controls allow adjusting chart presentation and show main parameters that are used for the buffer calculation along with their values.	Not available
Picture difference comparison	<i>Visual representation of picture difference at several processing stages based on a set of comparison criteria</i>	Allows picture comparison between decoded, predicted, unfiltered, residual, reference and memory stages and showing the difference in Compare, PSNR, PSNR clip, Subtraction, Temperature modes.	Not available
Memory	<i>Easy selection of a specific frame for further comparison against the current frame.</i>	The frames can be memorized at various processing stages: decoded, predicted, unfiltered, reference.	Not available
Overlays	<i>Quick configuration of the data display layout.</i>	Overlay Config The Panel is used for Overlay configuration and allows setting the color and display parameters to your liking.	Not available
		Slice boundaries: Tiles Slices <i>Customizable</i>	Shows all options together without customization. Is On by default.
		Partitions: Coded Block Prediction Transform Luma Transform Chroma <i>Customizable</i>	Shows all options together without customization. Is On by default.
		Motion vectors: Intra Angular Intra Plane Intra DC Inter Text <i>Customizable</i>	Shows all options together without customization. Is On by default. Text is not available
		Sizes: CTU CU PU TU Text <i>Customizable</i>	CU is On by default Text is not available
		Extended: Quant	Quant is On by default

		Quant Text SAO for HEVC CDEF, LR filtering for AV1 <i>Customizable</i>	
Command line tool	<i>Automatic statistics gathering for selected video arrays; customization of the configuration file</i>	• Analysis the whole array of video files automatically; • Saving of submitted results as a text file in CSV format for easy viewing in Excel; • Customization of a sequence of commands to perform your specific tasks; • Acceleration and automation of your regular tasks;	Not available
Applied filters	<i>Video display in the specified interlace mode or level of contrast/color richness.</i>	Interlace Interlace config HDR -> SDR (experimental)	Not available
Navigation	<i>Common navigation via the stream</i>	Play, Stop, Go to beginning, Go to end, Fast play, One frame forward, Jump to frame by frame type	
Search	<i>Quick positioning via the stream</i>	Search by stream/display order, time, offset.	